

```
In [1]: import numpy as np
import pandas as pd
```

```
In [3]: import os

for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
In [4]: import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as st

# This line is valid only in Jupyter Notebook
%matplotlib inline

sns.set(style="whitegrid")
```

```
In [6]: import warnings
warnings.filterwarnings('ignore')
```

```
In [7]: df = pd.read_csv(r"C:\Users\HAI\Downloads\heart.csv")
```

```
In [8]: df
```

```
Out[8]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tl
0	63	1	3	145	233	1	0	150	0	2.3	0	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	0	
3	56	1	1	120	236	0	1	178	0	0.8	2	0	
4	57	0	0	120	354	0	1	163	1	0.6	2	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	0	
299	45	1	3	110	264	0	1	132	0	1.2	1	0	
300	68	1	0	144	193	1	1	141	0	3.4	1	2	
301	57	1	0	130	131	0	1	115	1	1.2	1	1	
302	57	0	1	130	236	0	0	174	0	0.0	1	1	

303 rows × 14 columns



```
In [9]: # print the shape
print('The shape of the dataset : ', df.shape)
```

The shape of the dataset : (303, 14)

```
In [10]: df.head()
```

Out[10]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2



In [11]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         303 non-null    int64
1   sex         303 non-null    int64
2   cp          303 non-null    int64
3   trestbps    303 non-null    int64
4   chol        303 non-null    int64
5   fbs         303 non-null    int64
6   restecg     303 non-null    int64
7   thalach     303 non-null    int64
8   exang       303 non-null    int64
9   oldpeak     303 non-null    float64
10  slope       303 non-null    int64
11  ca          303 non-null    int64
12  thal        303 non-null    int64
13  target      303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [12]: `df.dtypes`

Out[12]:

```
age          int64
sex          int64
cp           int64
trestbps     int64
chol         int64
fbs          int64
restecg      int64
thalach      int64
exang        int64
oldpeak      float64
slope        int64
ca           int64
thal         int64
target       int64
dtype: object
```

In [13]: `df.describe()`

Out[13]:

	age	sex	cp	trestbps	chol	fbs	restecg
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528000
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525000
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

In [14]: `df.columns`

Out[14]: Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'], dtype='object')

In [15]: `df['target'].nunique()`

Out[15]: 2

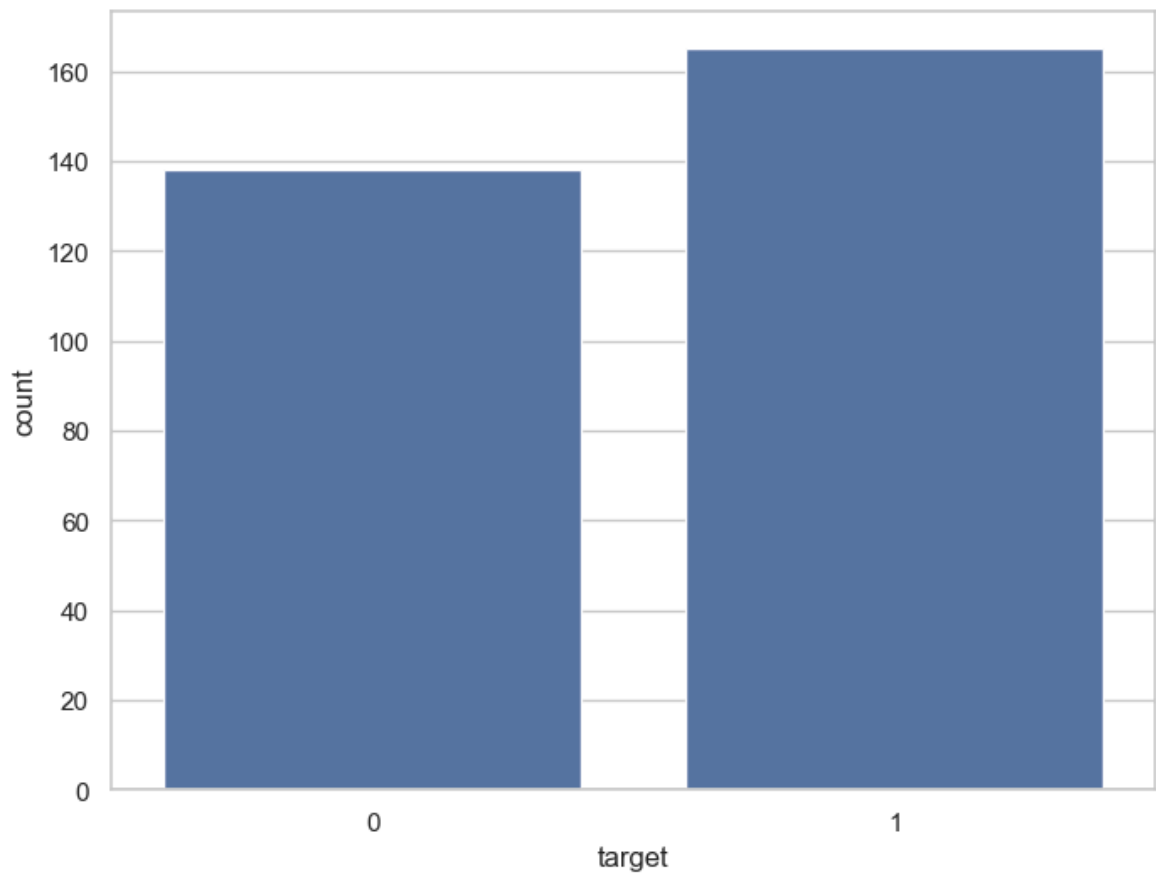
In [16]: `df['target'].unique()`

Out[16]: array([1, 0], dtype=int64)

In [17]: `df['target'].value_counts()`

Out[17]: target
 1 165
 0 138
 Name: count, dtype: int64

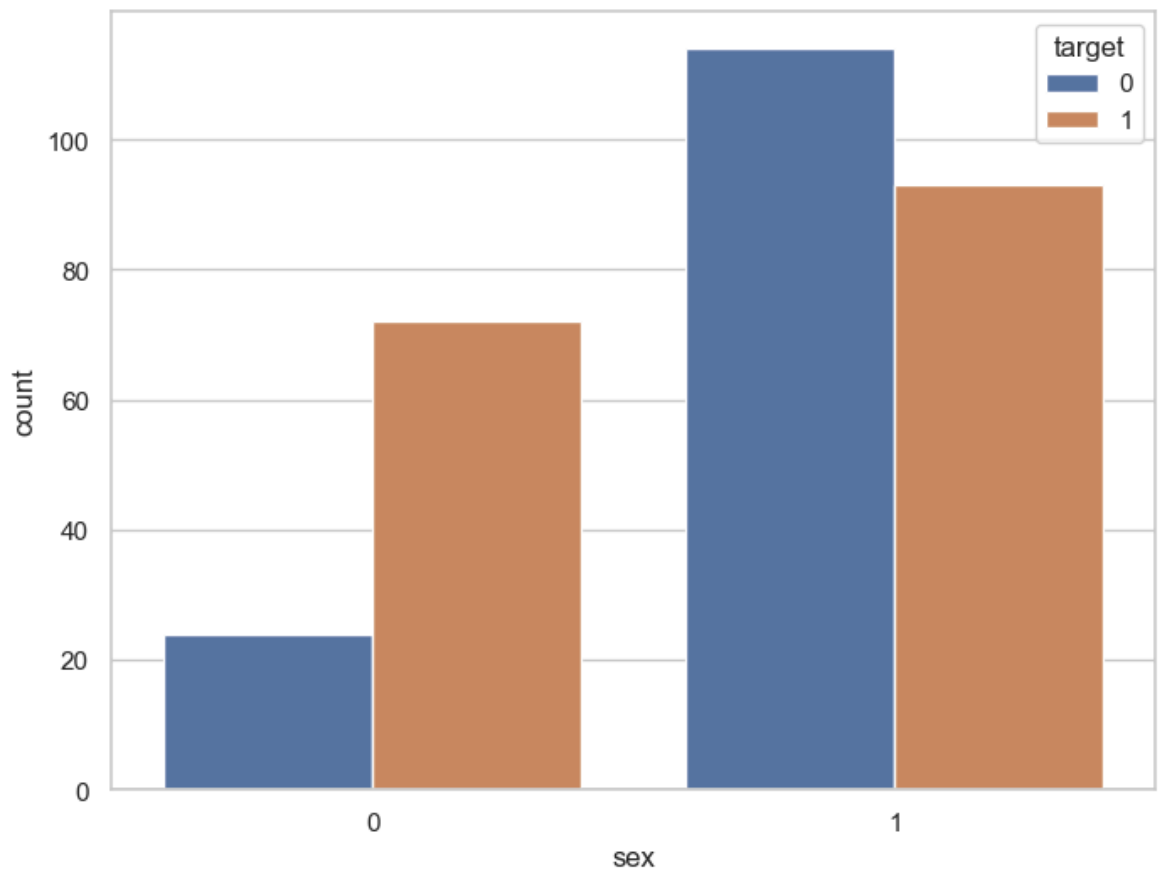
In [19]: `f, ax = plt.subplots(figsize=(8, 6))
 ax = sns.countplot(x="target", data=df)
 plt.show()`



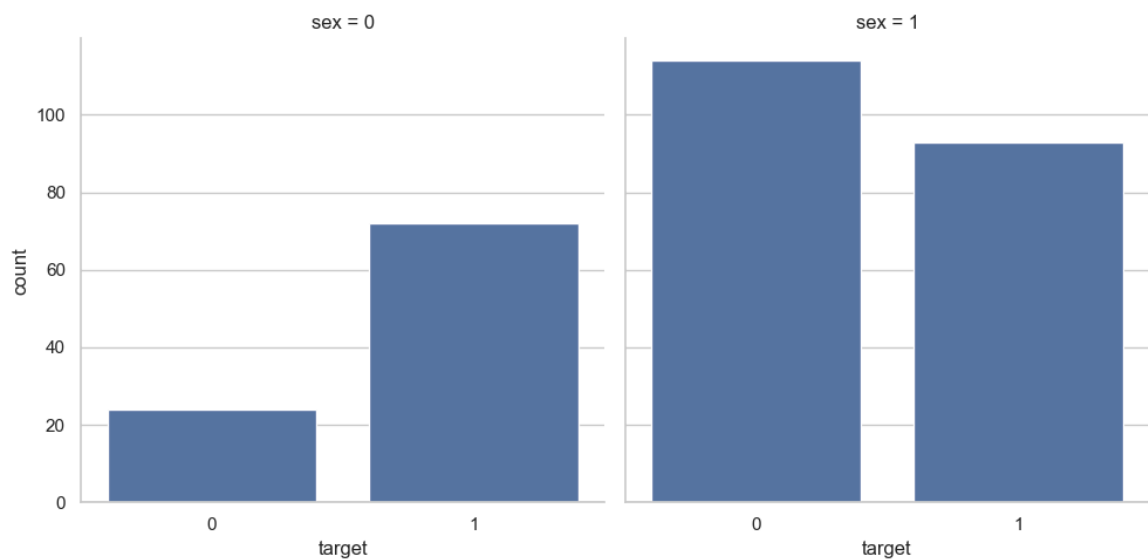
```
In [20]: df.groupby('sex')['target'].value_counts()
```

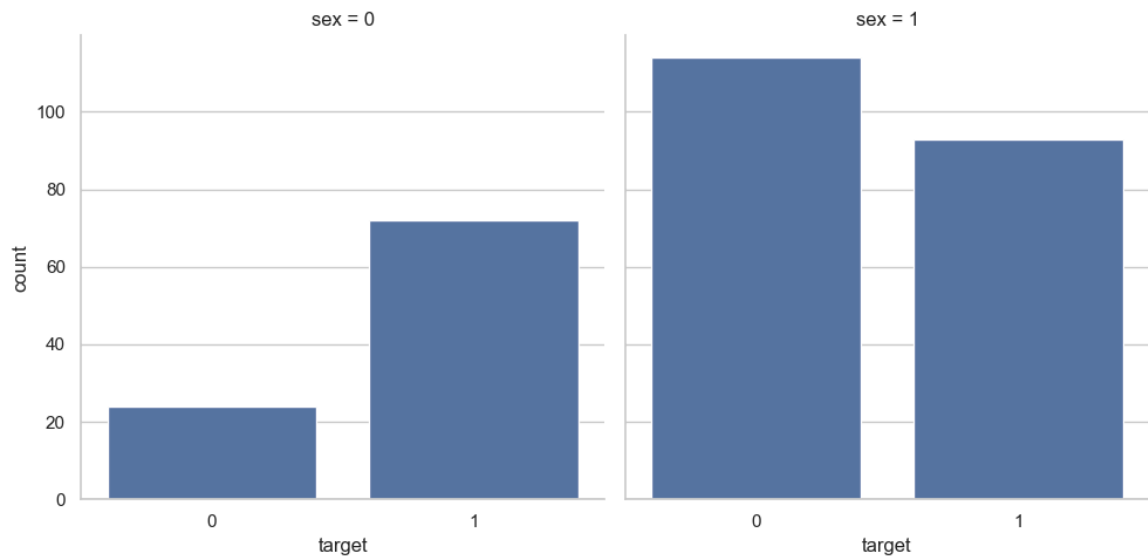
```
Out[20]: sex  target
0      1      72
      0      24
1      0     114
      1      93
Name: count, dtype: int64
```

```
In [21]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="sex", hue="target", data=df)
plt.show()
```

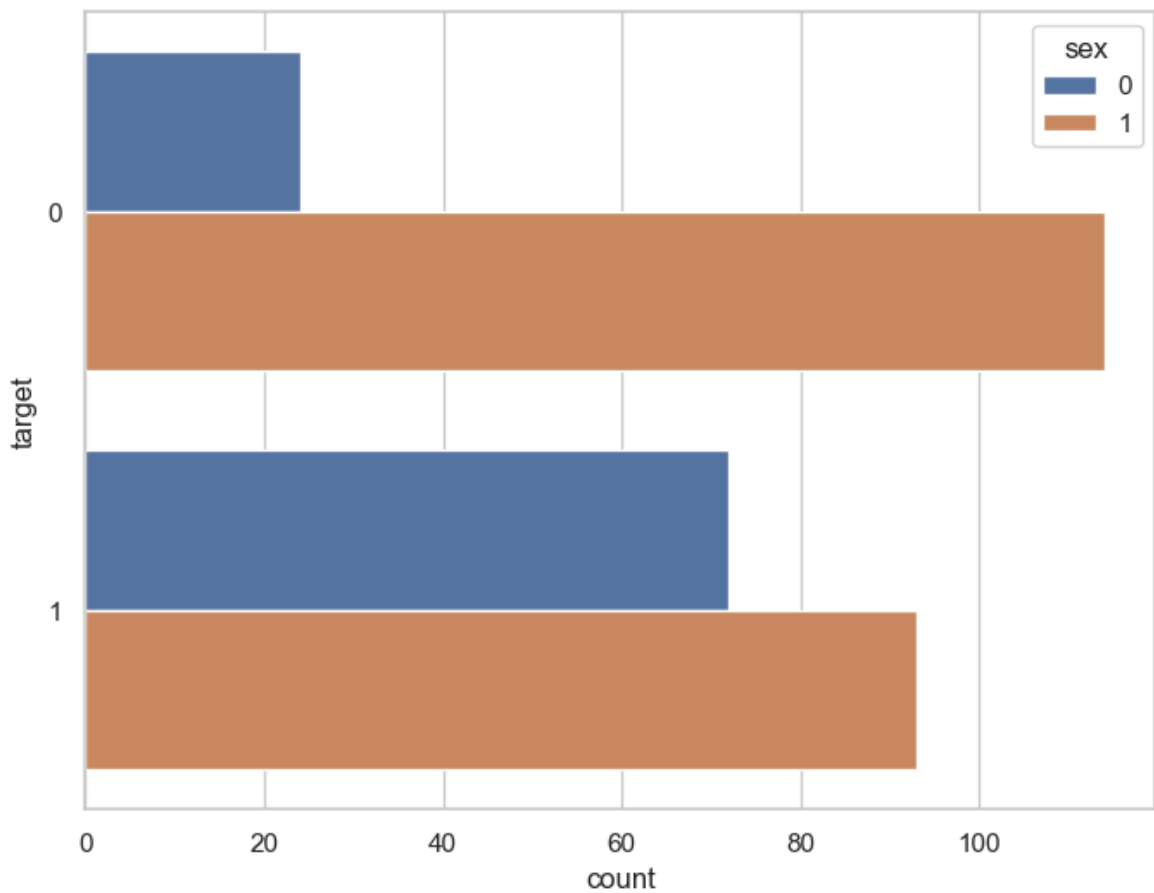


```
In [24]: ax = sns.catplot(x="target", col="sex", data=df, kind="count", height=5, aspect=
plt.show()
```

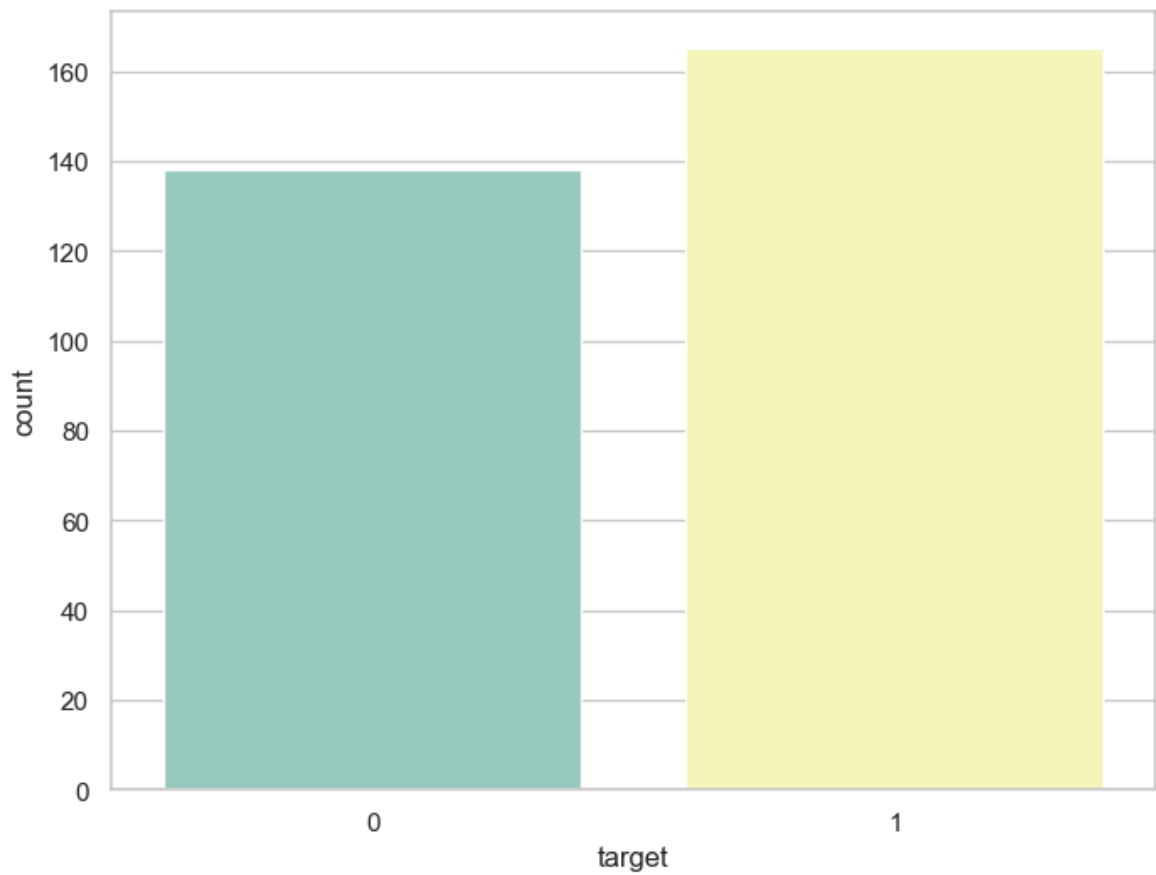




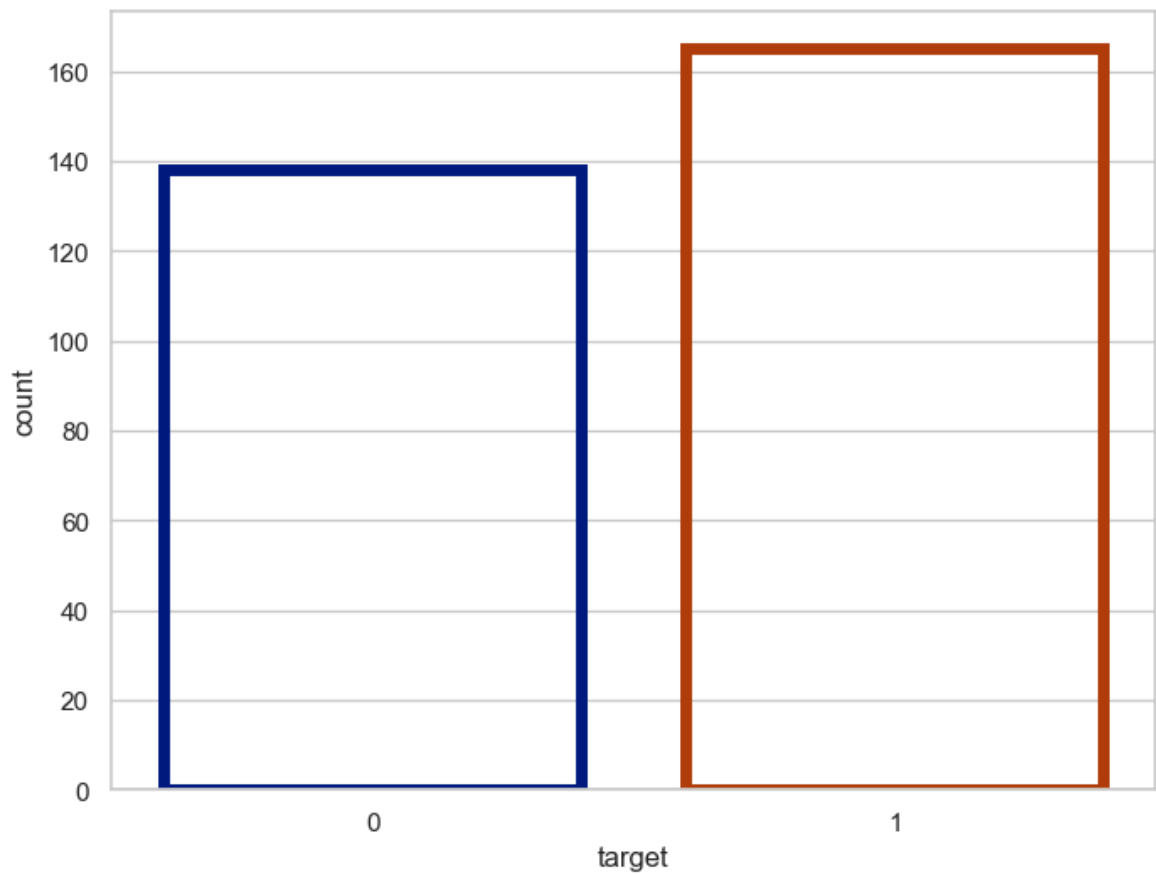
```
In [27]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(y="target", hue="sex", data=df)
plt.show()
```



```
In [28]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, palette="Set3")
plt.show()
```

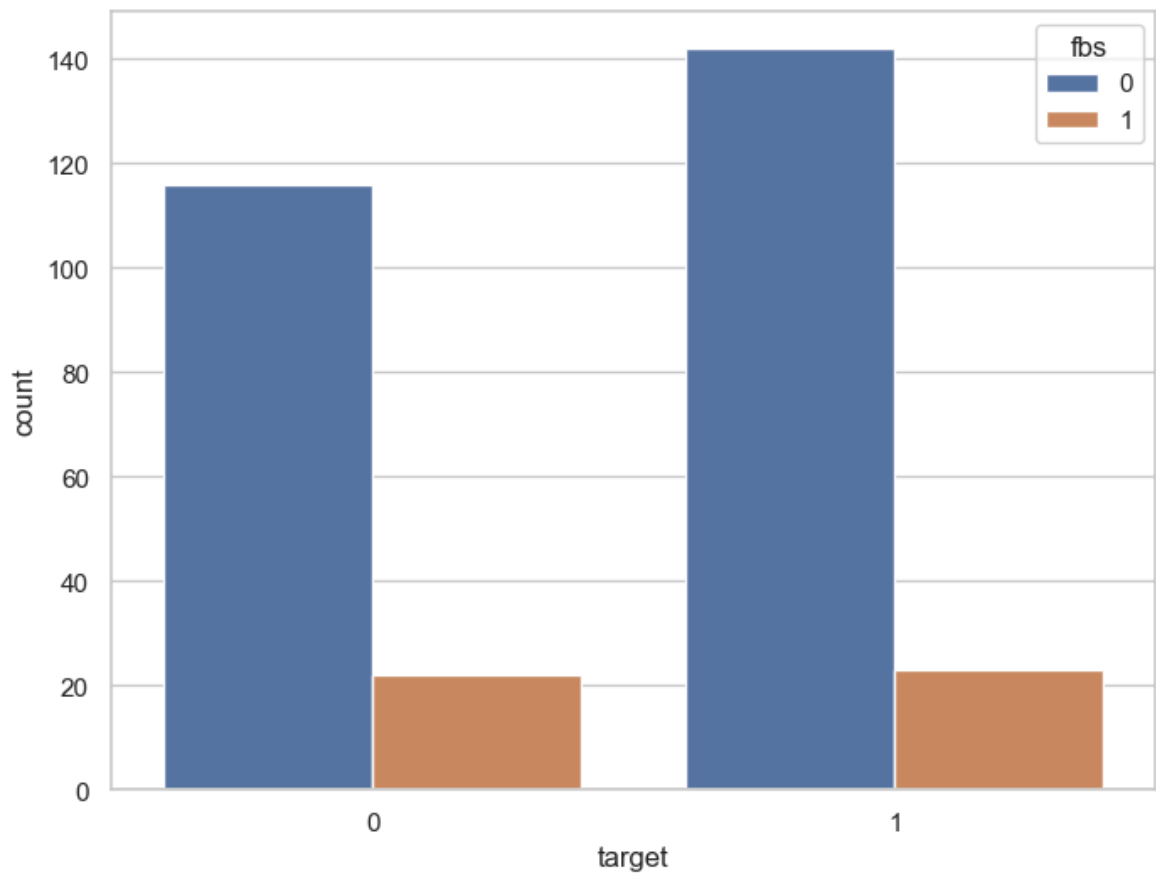


```
In [29]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", data=df, facecolor=(0, 0, 0, 0), linewidth=5, edge  
plt.show()
```

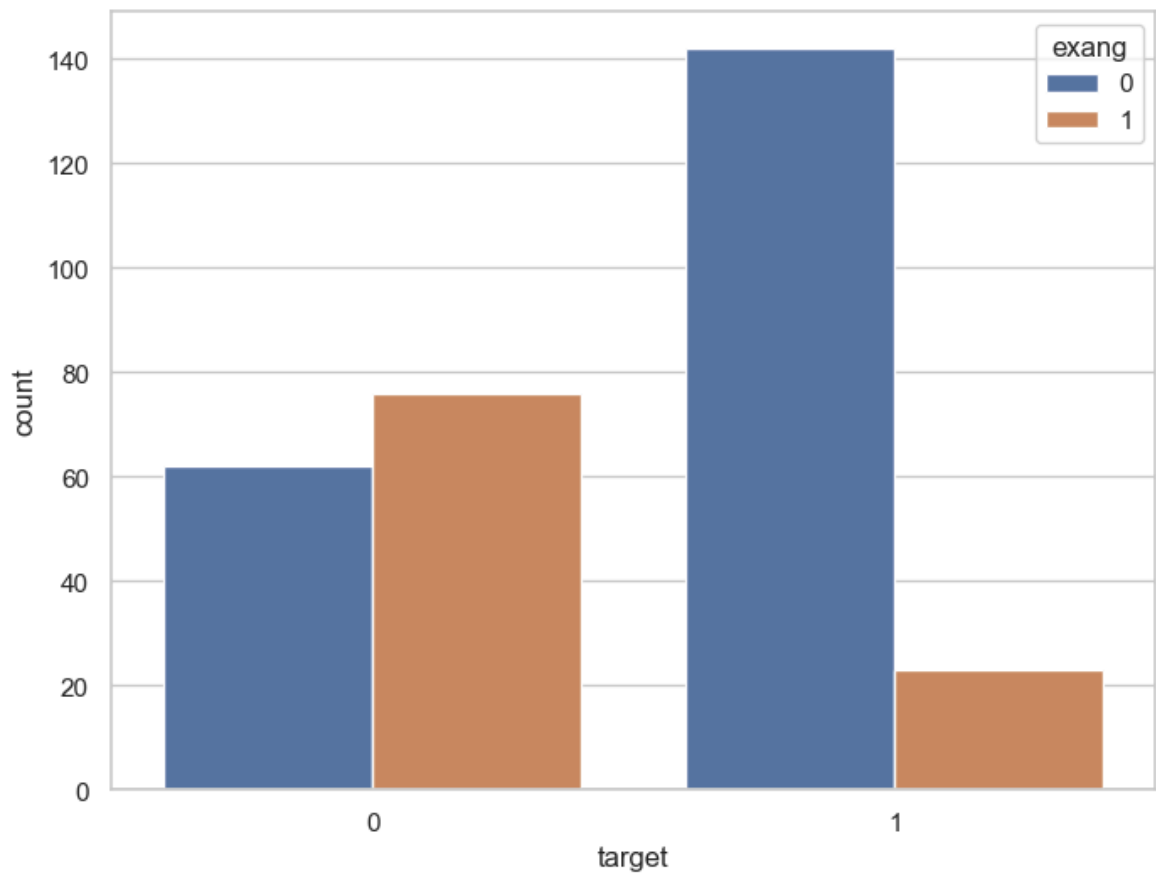


```
In [30]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", hue="fbs", data=df)
```

```
plt.show()
```



```
In [31]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", hue="exang", data=df)  
plt.show()
```




```
In [32]: correlation = df.corr()
```

```
In [33]: correlation['target'].sort_values(ascending=False)
```

```
Out[33]: target      1.000000  
         cp         0.433798  
         thalach    0.421741  
         slope      0.345877  
         restecg    0.137230  
         fbs        -0.028046  
         chol       -0.085239  
         trestbps   -0.144931  
         age        -0.225439  
         sex        -0.280937  
         thal       -0.344029  
         ca         -0.391724  
         oldpeak    -0.430696  
         exang      -0.436757  
         Name: target, dtype: float64
```

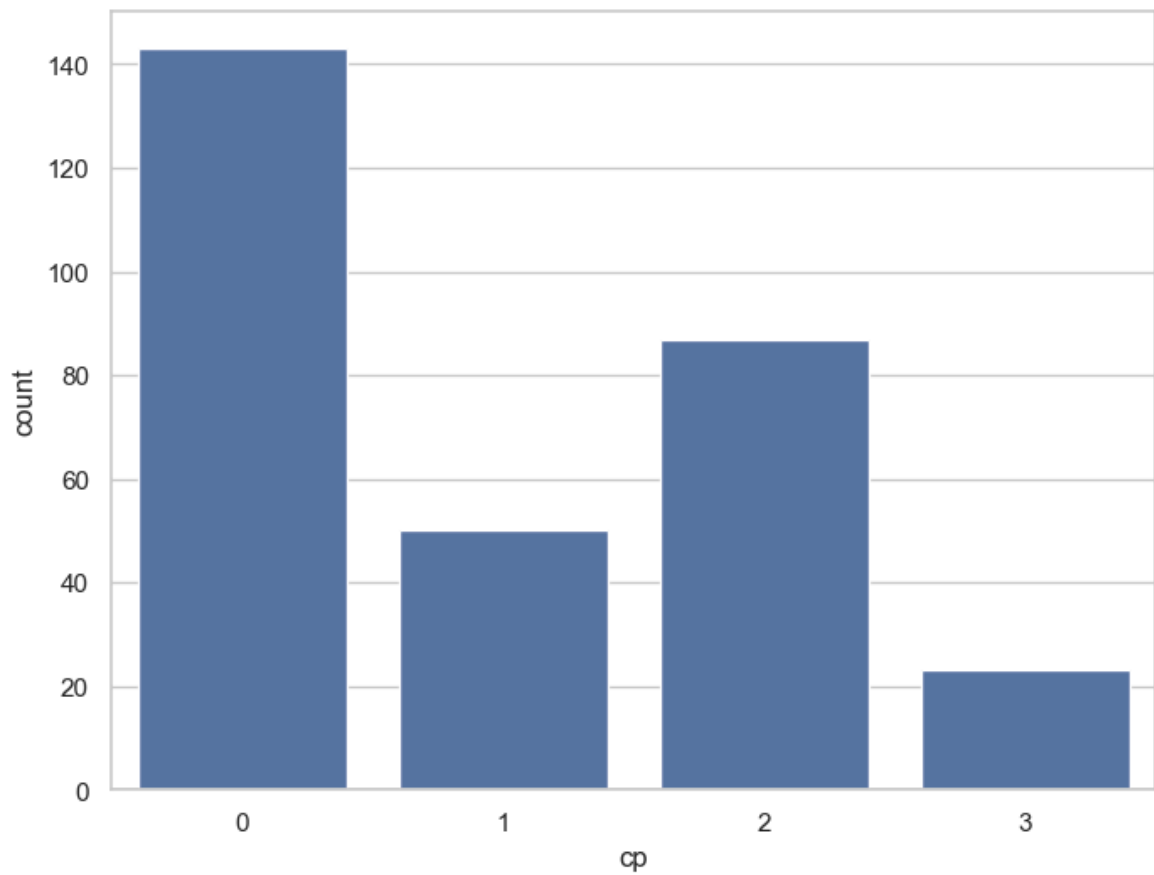
```
In [34]: df['cp'].nunique()
```

```
Out[34]: 4
```

```
In [35]: df['cp'].value_counts()
```

```
Out[35]: cp  
0      143  
2       87  
1       50  
3       23  
         Name: count, dtype: int64
```

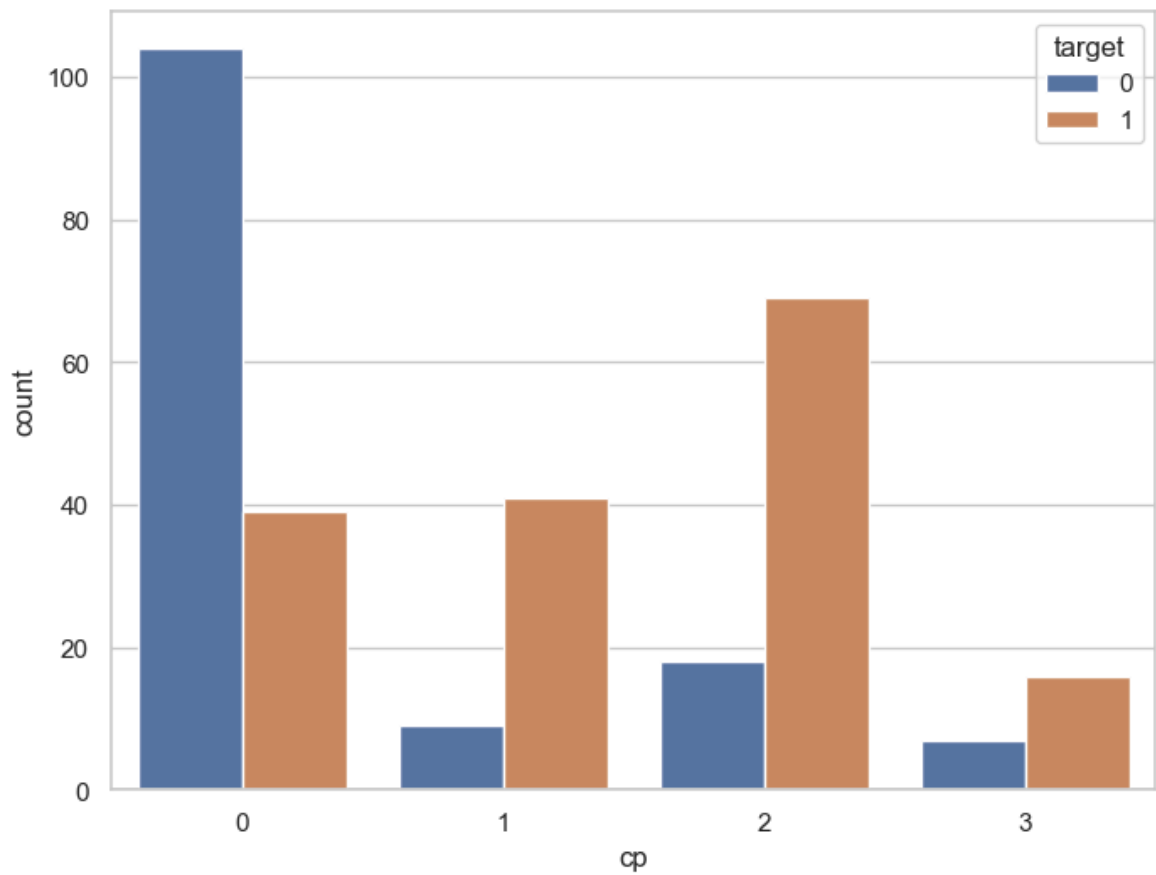
```
In [36]: f, ax = plt.subplots(figsize=(8, 6))  
         ax = sns.countplot(x="cp", data=df)  
         plt.show()
```



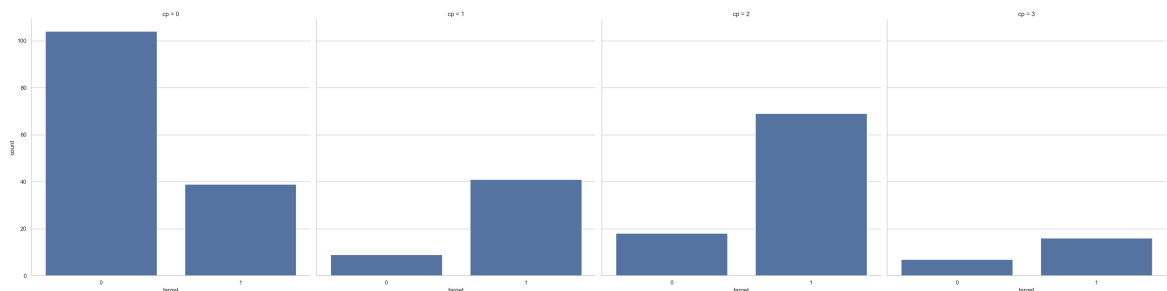
```
In [37]: df.groupby('cp')['target'].value_counts()
```

```
Out[37]: cp target
0  0      104
   1       39
1  1       41
   0        9
2  1       69
   0       18
3  1       16
   0        7
Name: count, dtype: int64
```

```
In [38]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="cp", hue="target", data=df)
plt.show()
```



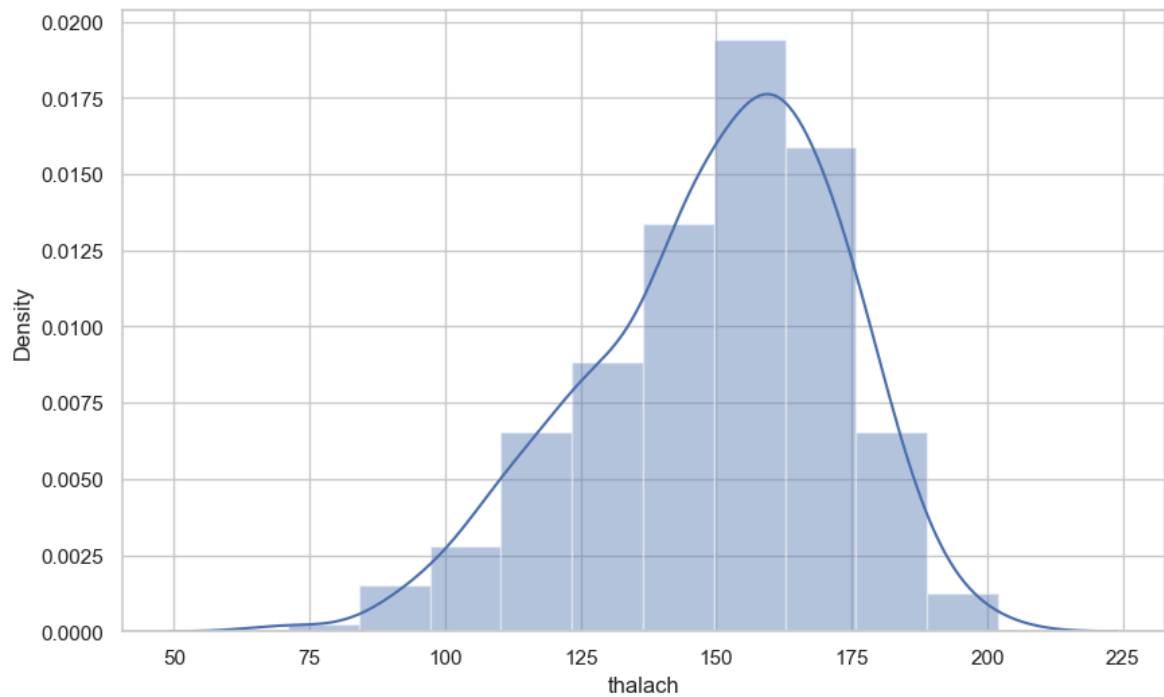
```
In [39]: ax = sns.catplot(x="target", col="cp", data=df, kind="count", height=8, aspect=1)
plt.show()
```



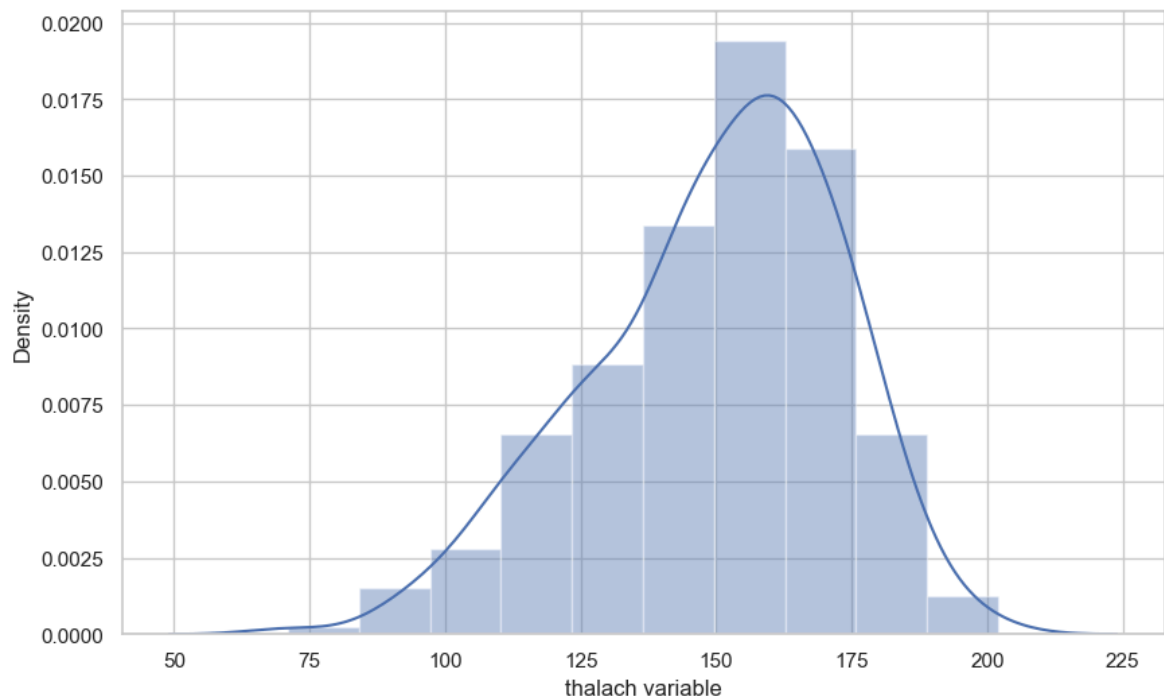
```
In [40]: df['thalach'].nunique()
```

```
Out[40]: 91
```

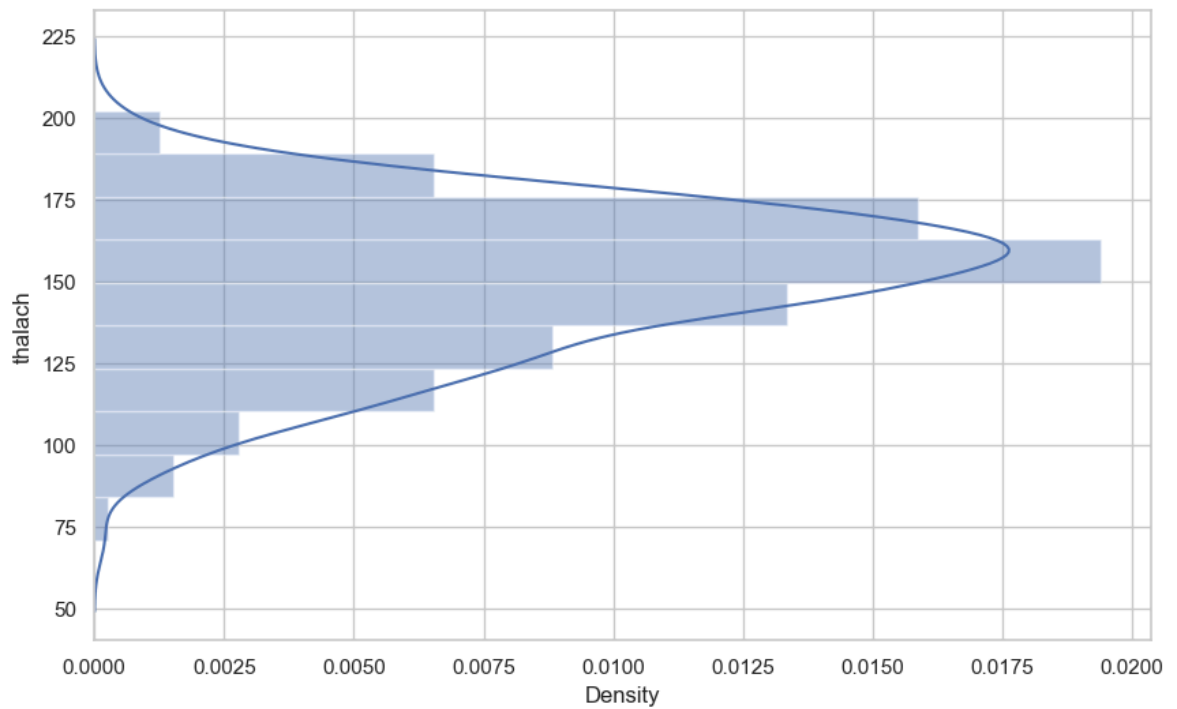
```
In [41]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10)
plt.show()
```



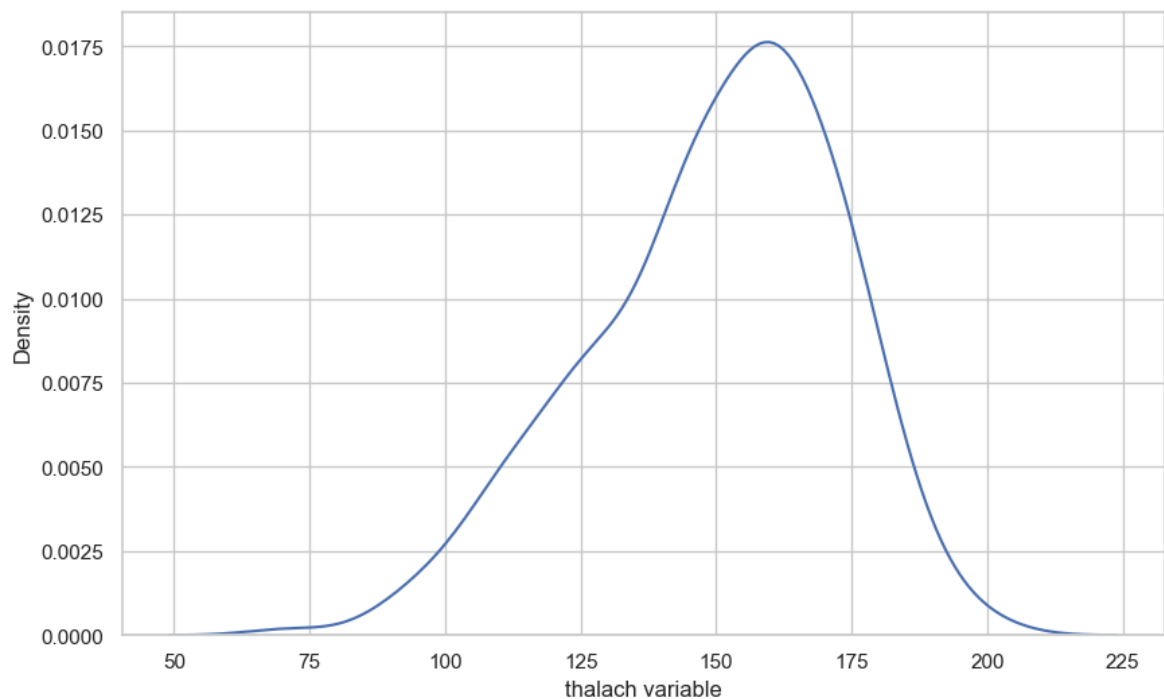
```
In [42]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.distplot(x, bins=10)
plt.show()
```



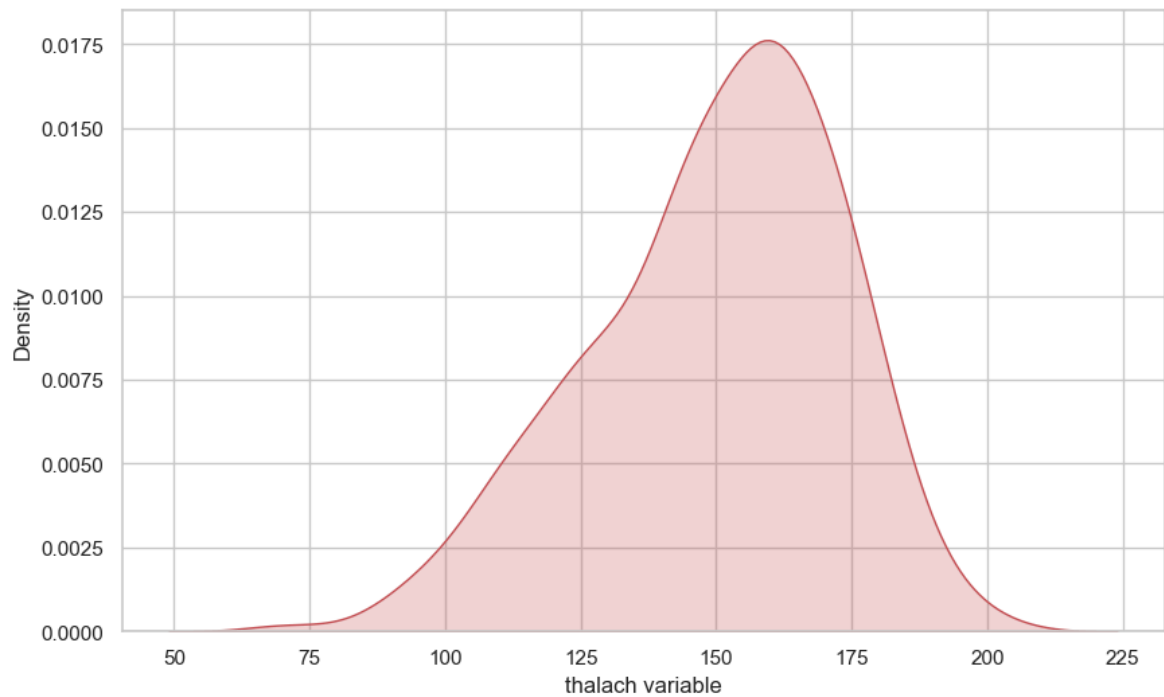
```
In [43]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10, vertical=True)
plt.show()
```



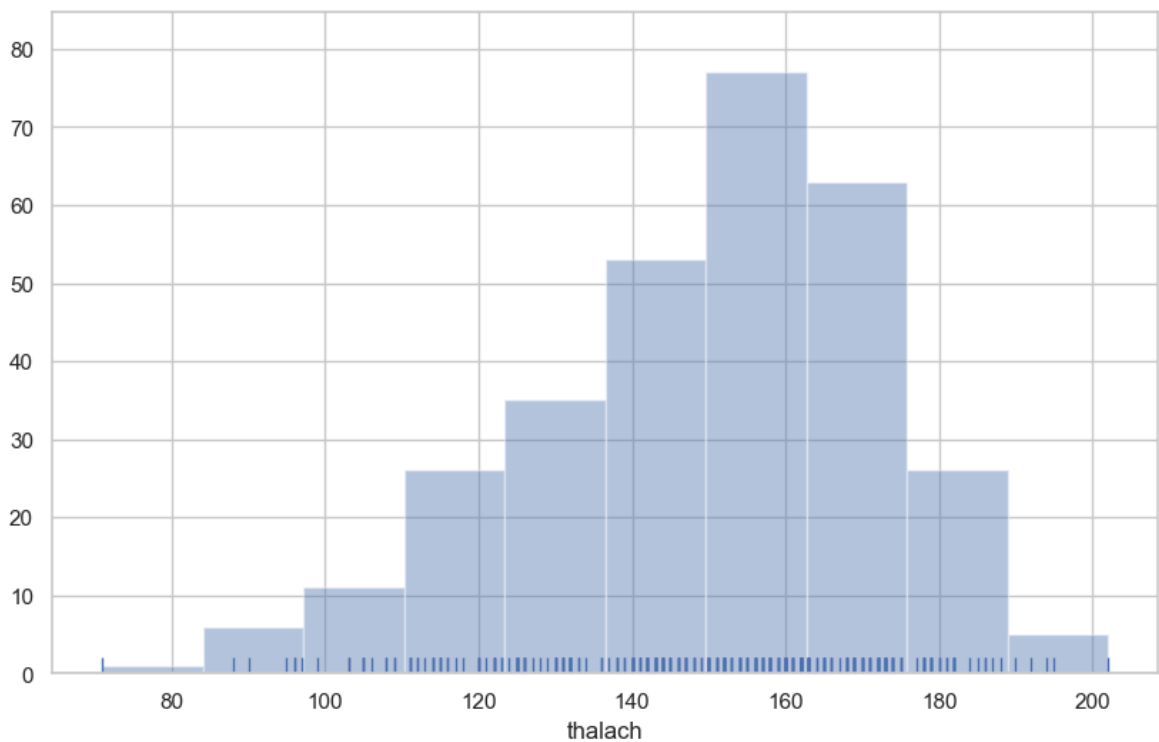
```
In [44]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x)
plt.show()
```



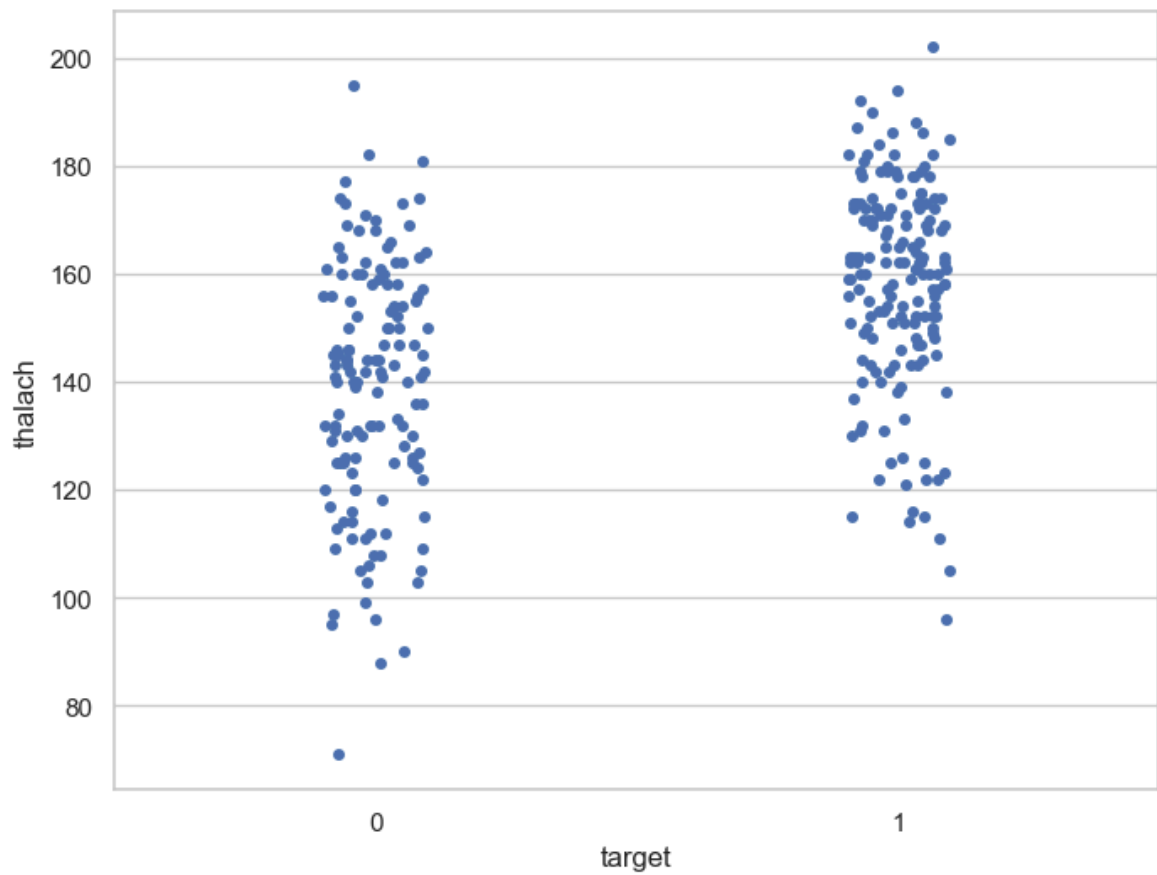
```
In [45]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x, shade=True, color='r')
plt.show()
```



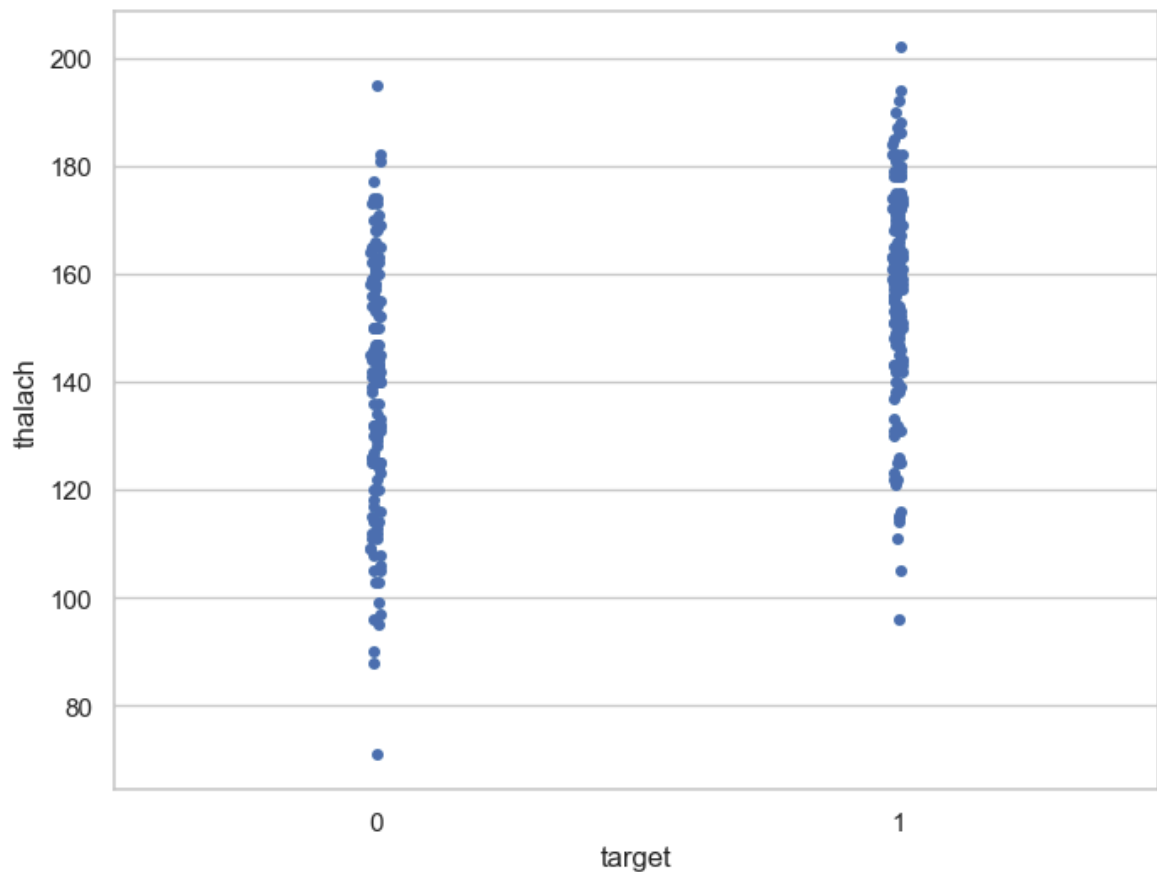
```
In [46]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, kde=False, rug=True, bins=10)
plt.show()
```



```
In [47]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="thalach", data=df)
plt.show()
```

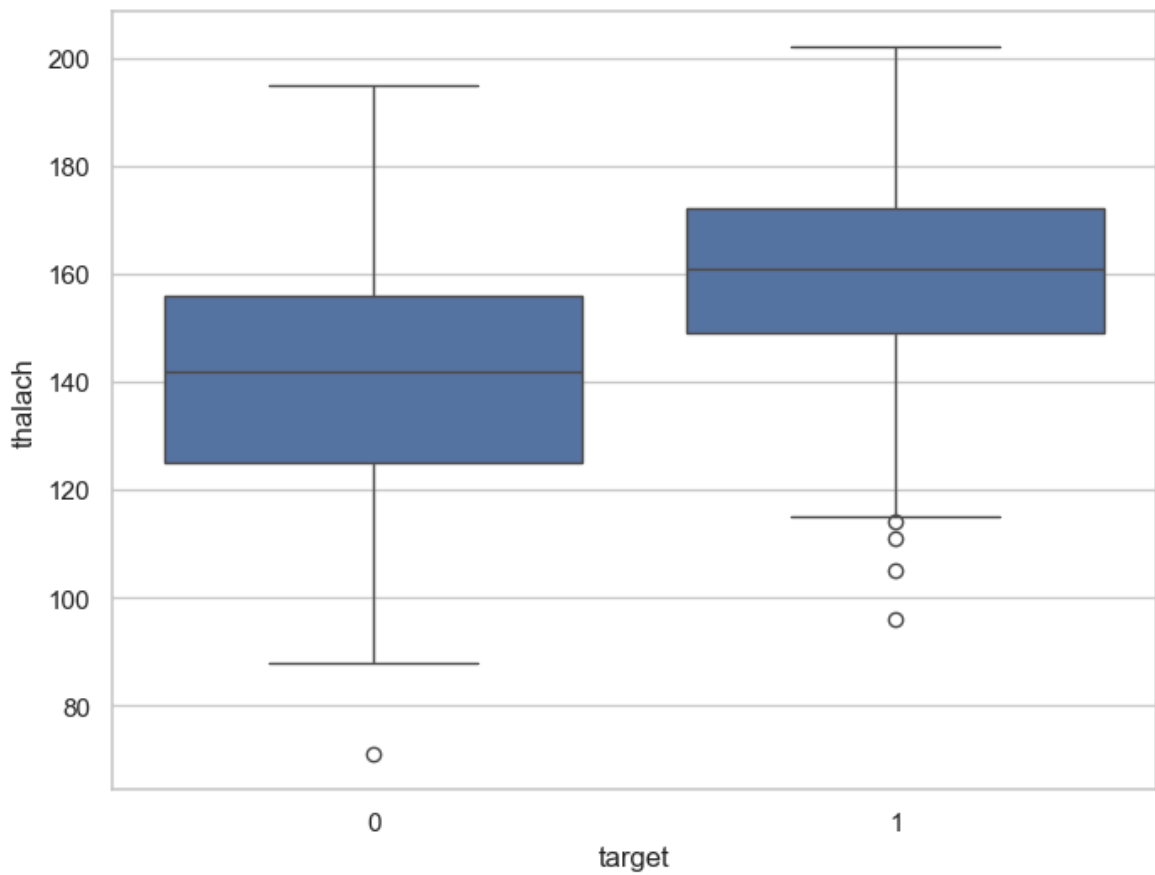


```
In [48]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="thalach", data=df, jitter = 0.01)
plt.show()
```

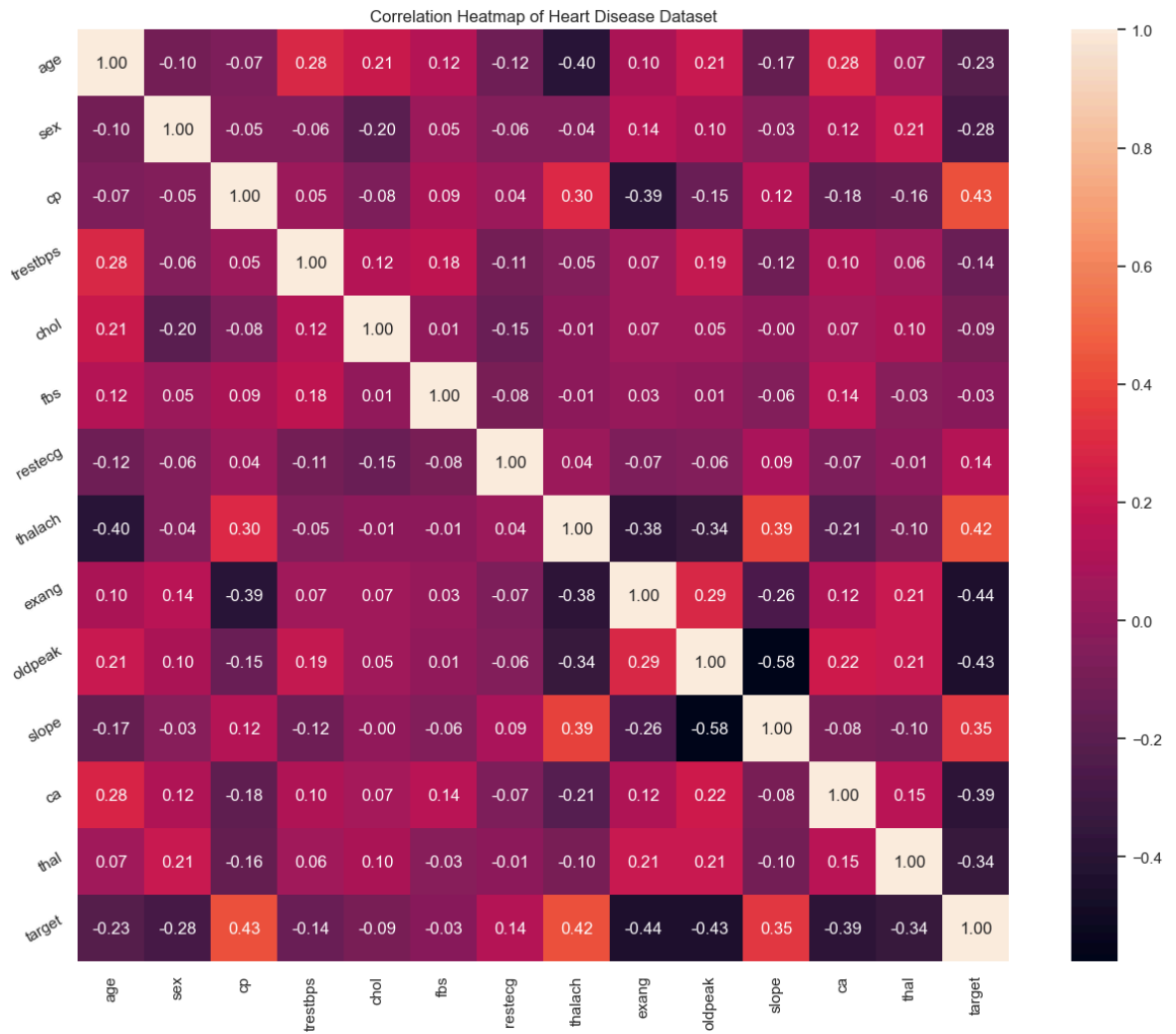


```
In [49]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="target", y="thalach", data=df)
```

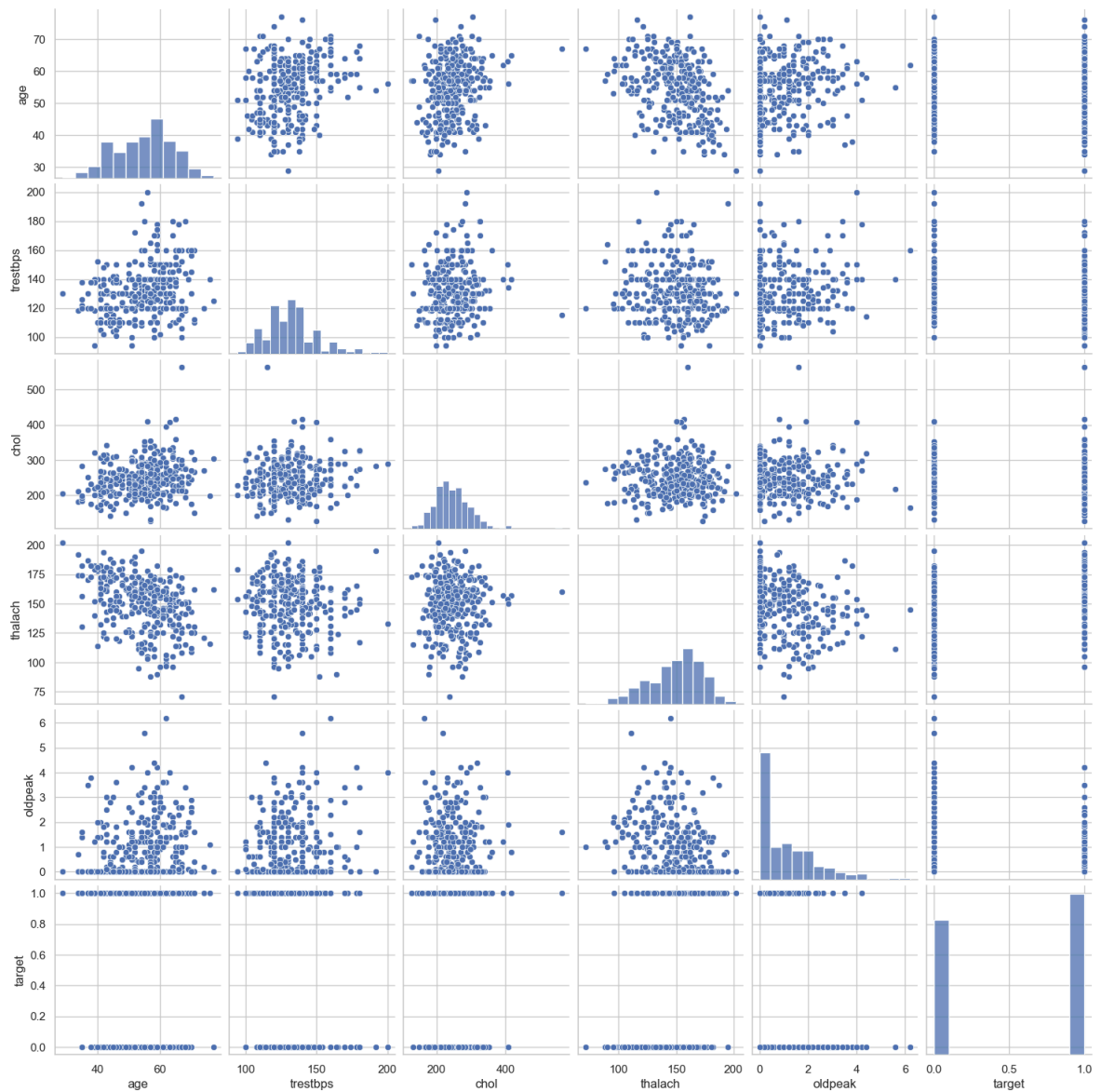
```
plt.show()
```



```
In [50]: plt.figure(figsize=(16,12))
plt.title('Correlation Heatmap of Heart Disease Dataset')
a = sns.heatmap(correlation, square=True, annot=True, fmt='.2f', linecolor='white')
a.set_xticklabels(a.get_xticklabels(), rotation=90)
a.set_yticklabels(a.get_yticklabels(), rotation=30)
plt.show()
```

```
In [52]: num_var = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'target' ]
sns.pairplot(df[num_var], kind='scatter', diag_kind='hist')
plt.show()
```



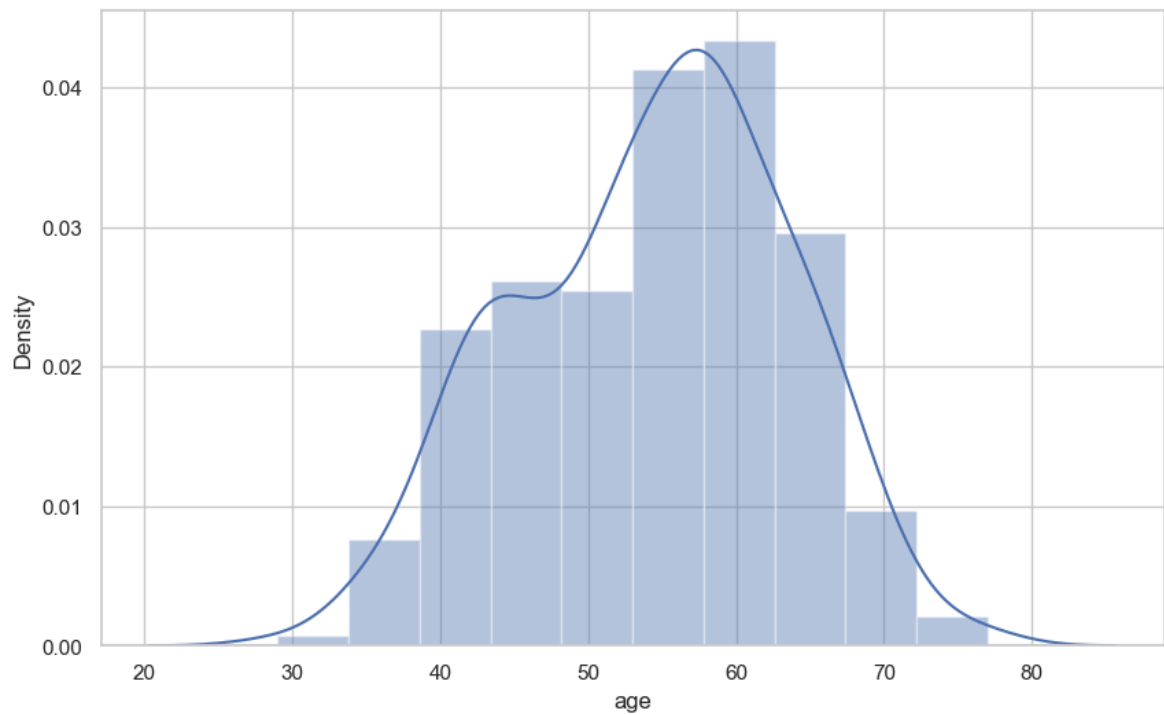
```
In [53]: df['age'].unique()
```

```
Out[53]: 41
```

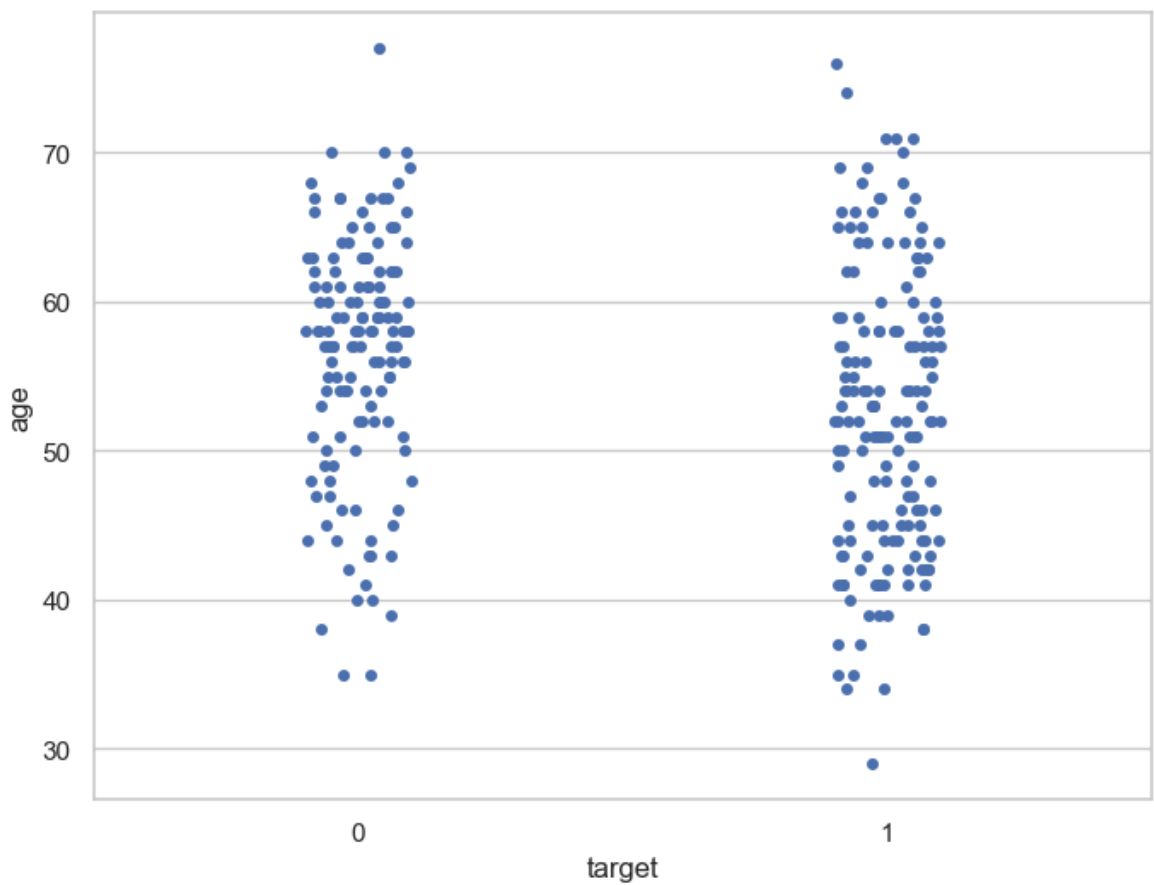
```
In [54]: df['age'].describe()
```

```
Out[54]: count    303.000000
mean       54.366337
std        9.082101
min        29.000000
25%        47.500000
50%        55.000000
75%        61.000000
max        77.000000
Name: age, dtype: float64
```

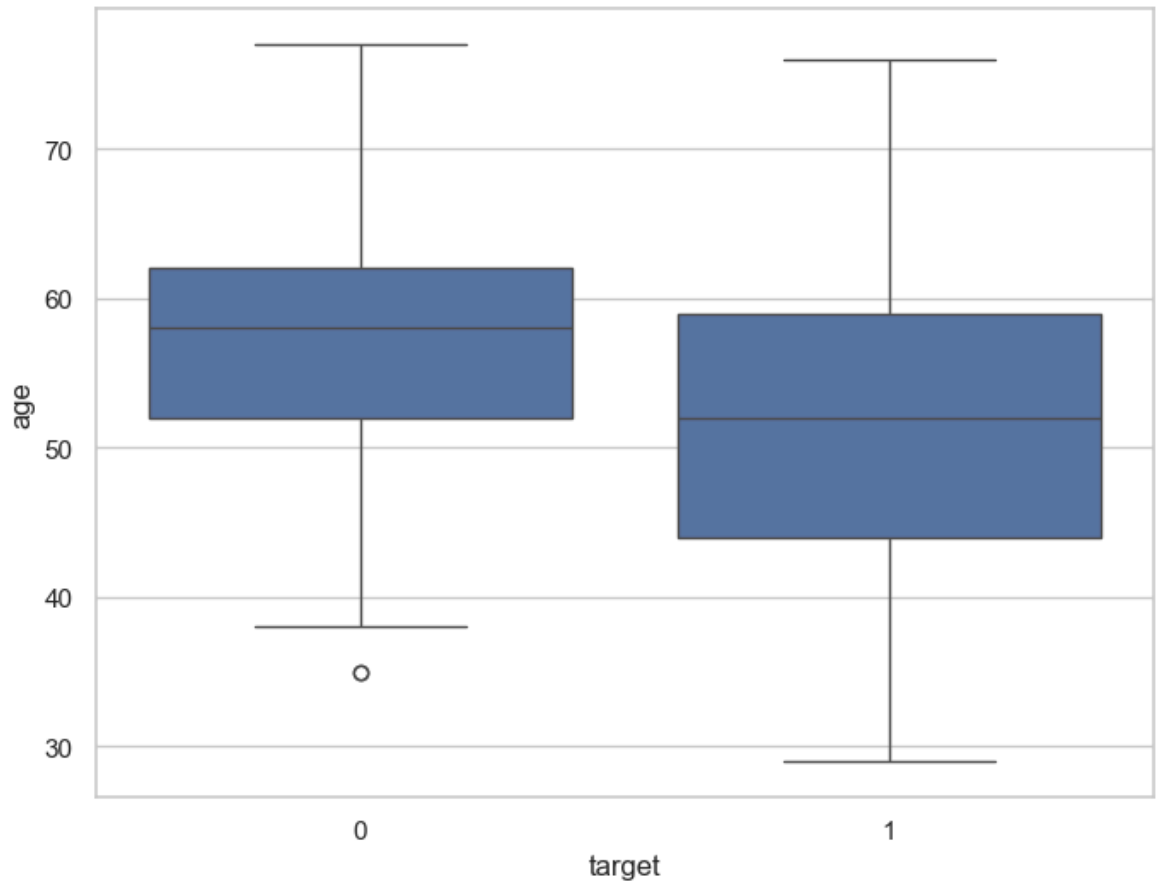
```
In [55]: f, ax = plt.subplots(figsize=(10,6))
x = df['age']
ax = sns.distplot(x, bins=10)
plt.show()
```



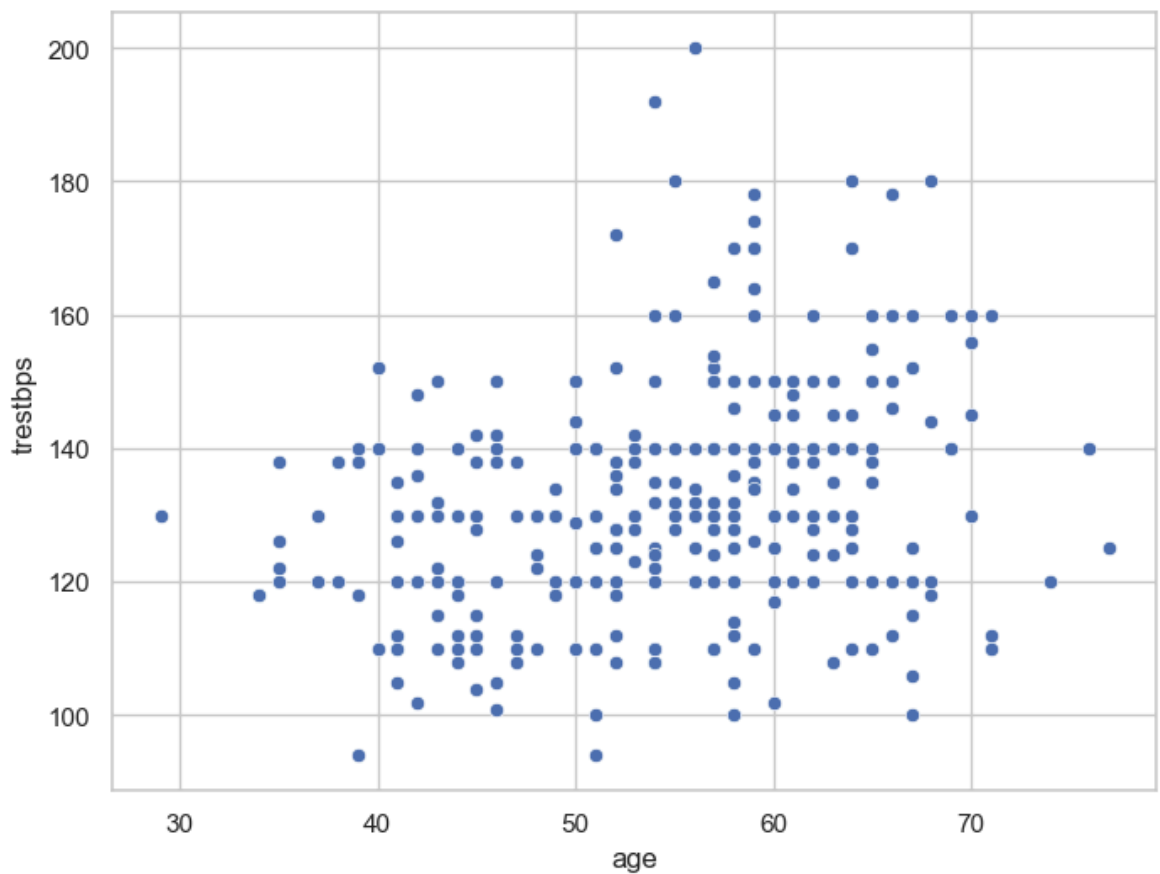
```
In [56]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="age", data=df)
plt.show()
```



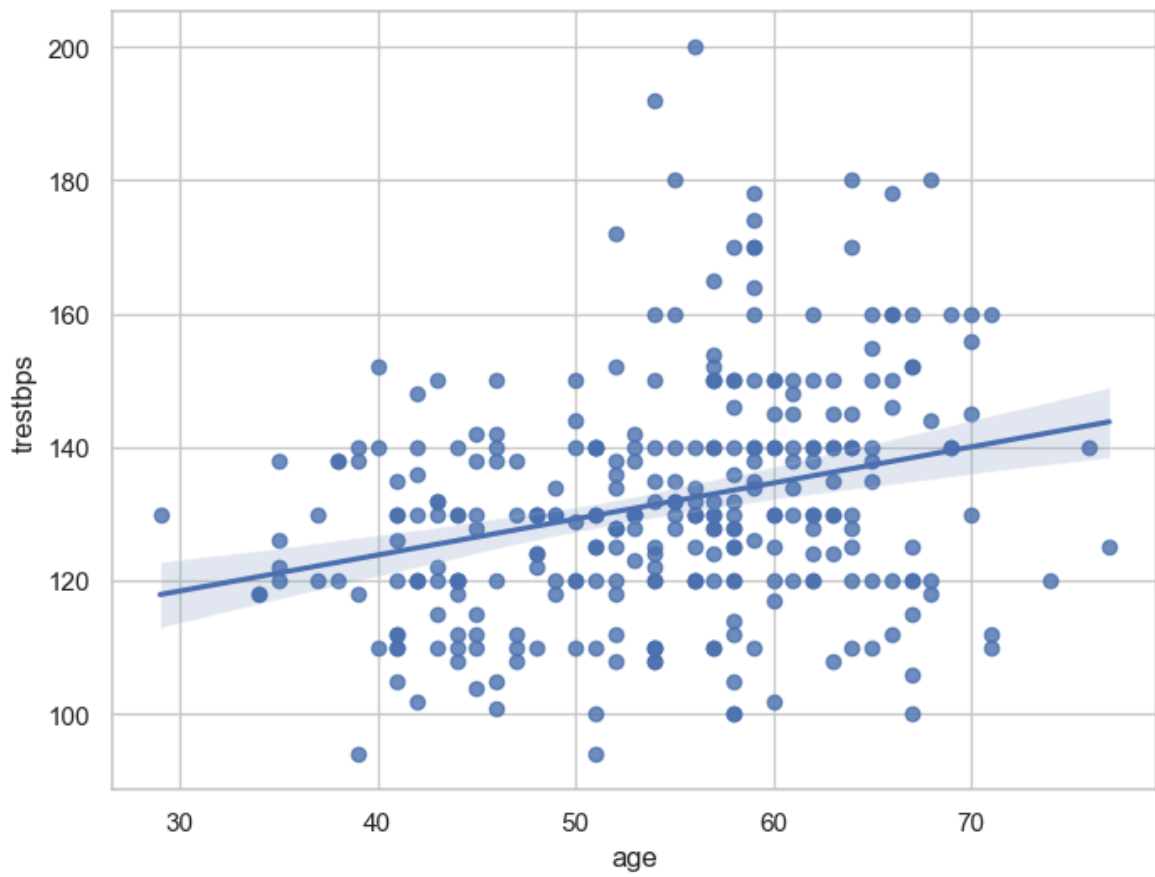
```
In [57]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="target", y="age", data=df)
plt.show()
```



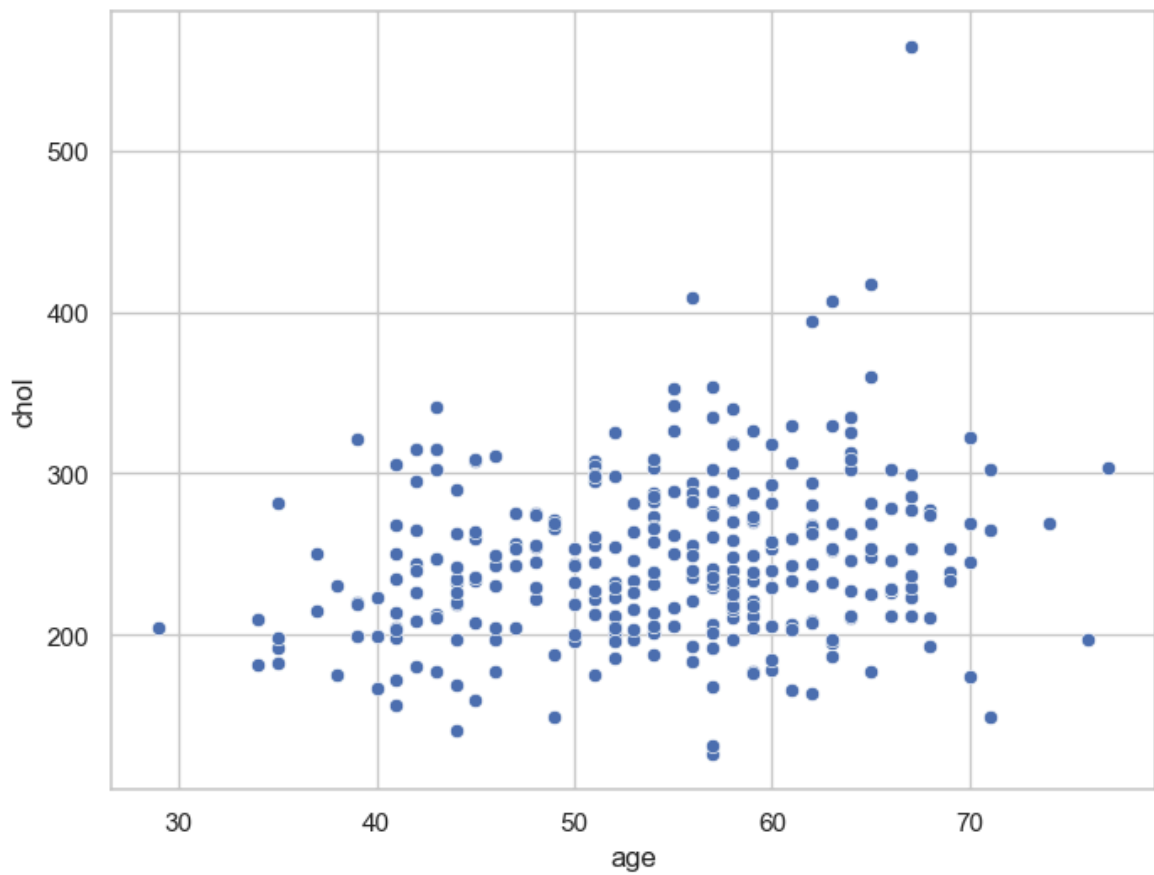
```
In [58]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="age", y="trestbps", data=df)
plt.show()
```



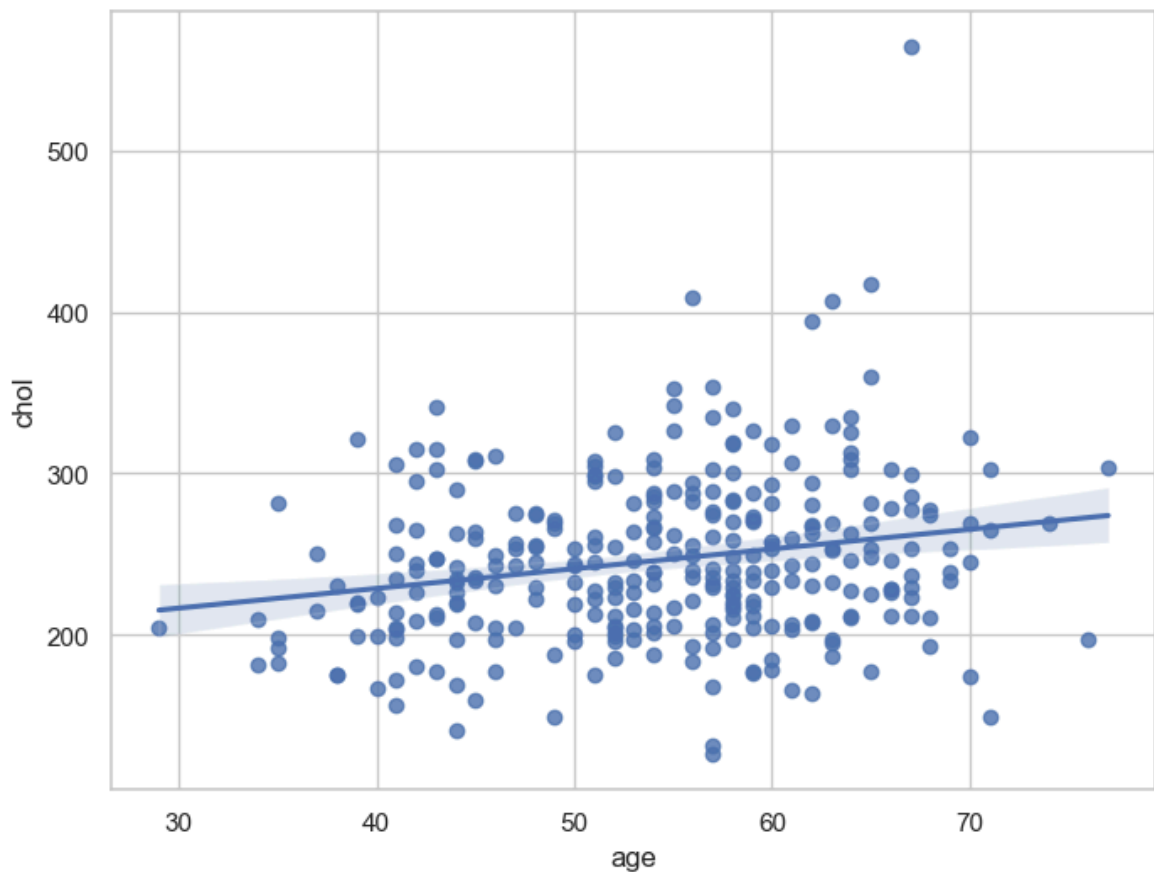
```
In [59]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="age", y="trestbps", data=df)  
plt.show()
```



```
In [60]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.scatterplot(x="age", y="chol", data=df)  
plt.show()
```

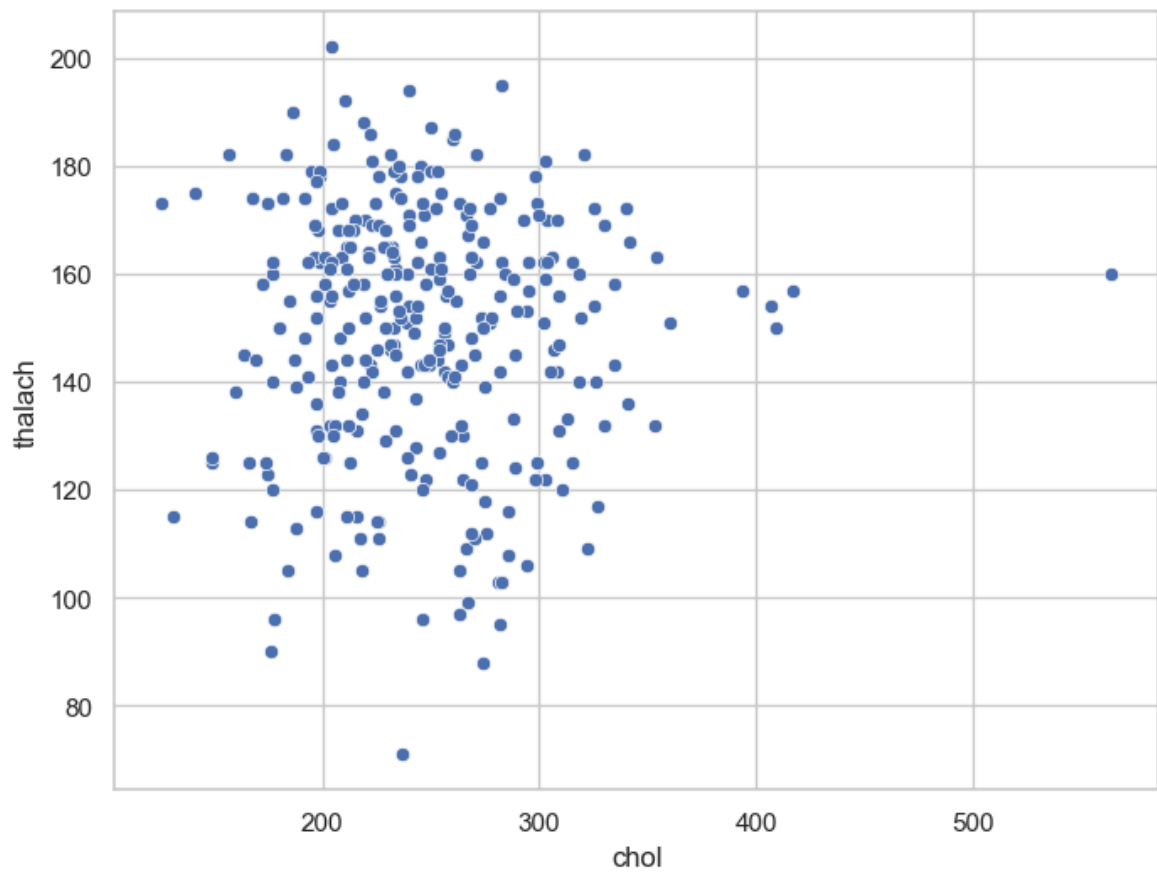


```
In [61]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="age", y="chol", data=df)
plt.show()
```

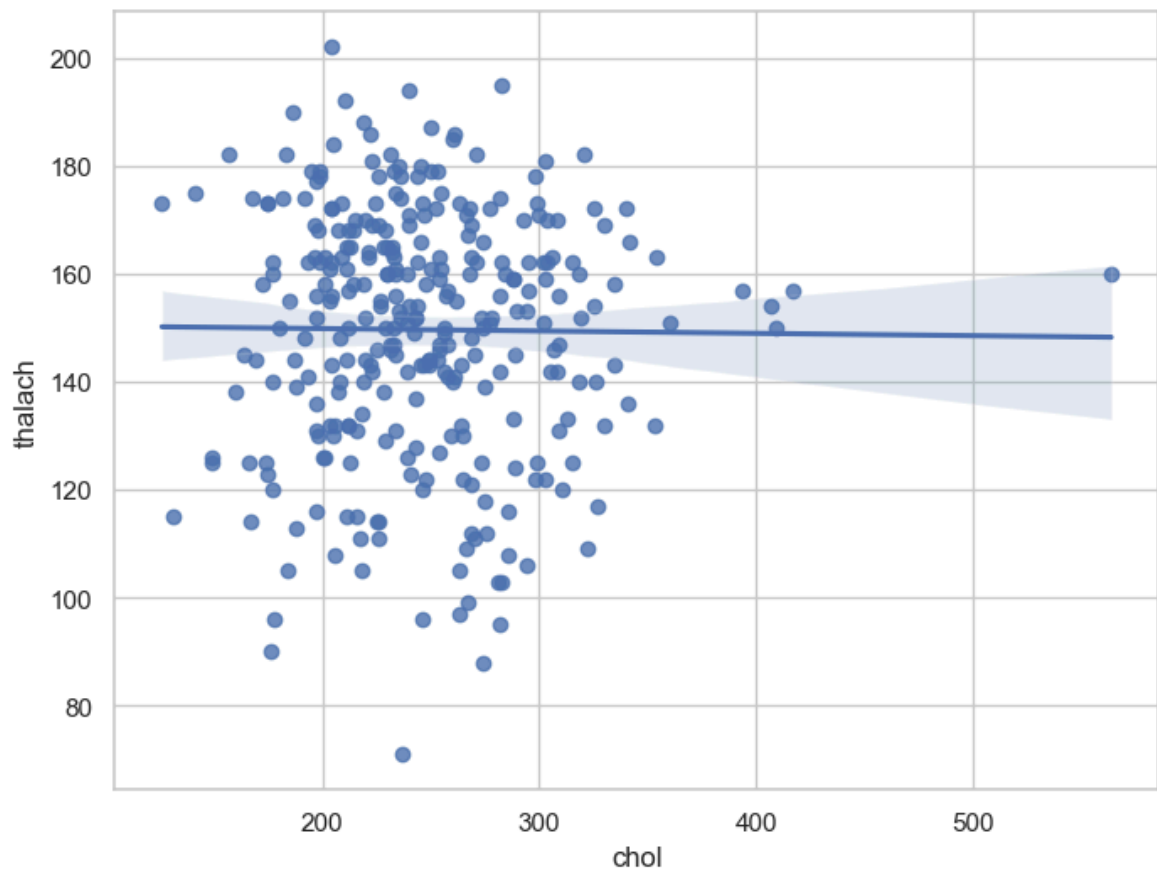


```
In [62]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="chol", y = "thalach", data=df)
```

```
plt.show()
```



```
In [63]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="chol", y="thalach", data=df)  
plt.show()
```



In [64]: *# check for missing values*

```
df.isnull().sum()
```

Out[64]:

age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

In [65]: *#assert that there are no missing values in the dataframe*

```
assert pd.notnull(df).all().all()
```

In [66]: *#assert all values are greater than or equal to 0*

```
assert (df >= 0).all().all()
```

In [67]: df['age'].describe()

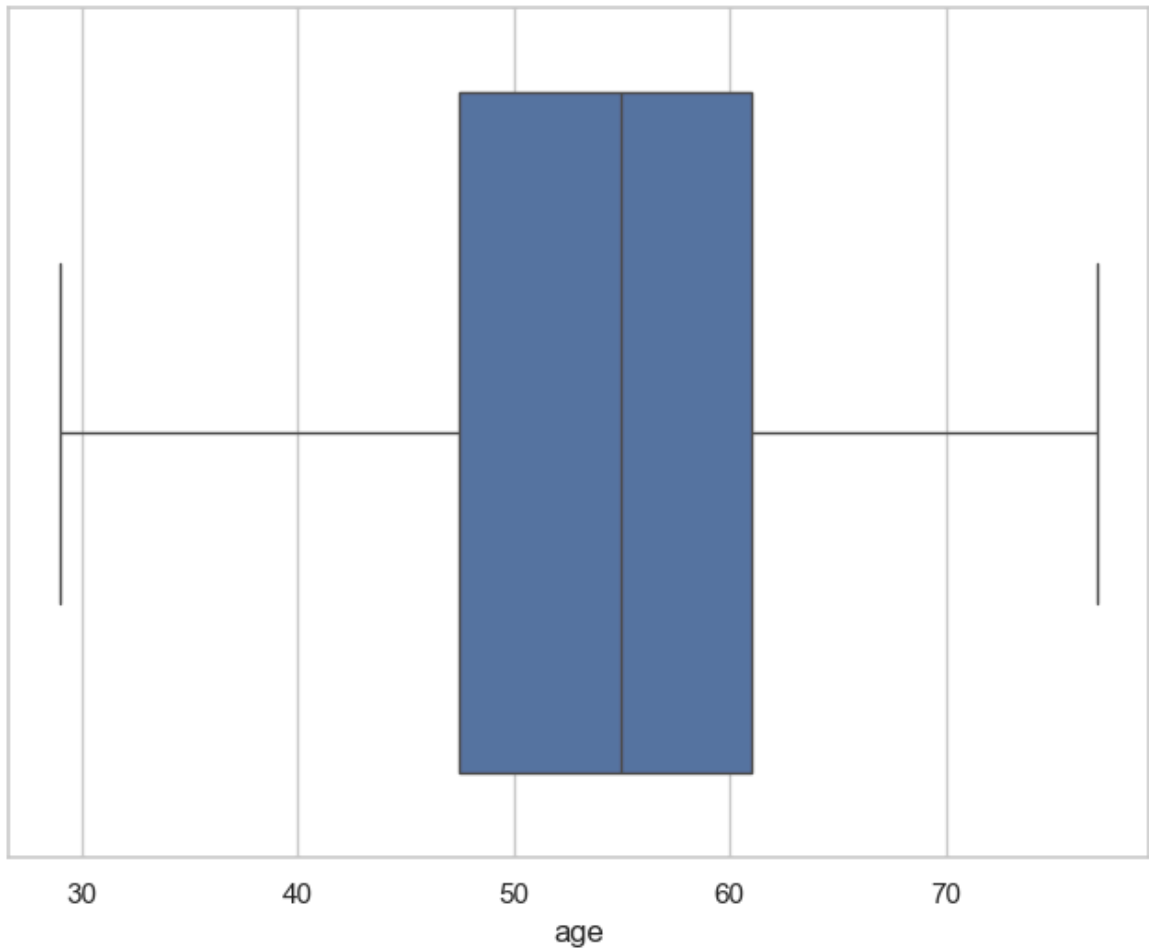
Out[67]:

count	303.000000
mean	54.366337
std	9.082101
min	29.000000
25%	47.500000
50%	55.000000
75%	61.000000
max	77.000000

Name: age, dtype: float64

In [68]:

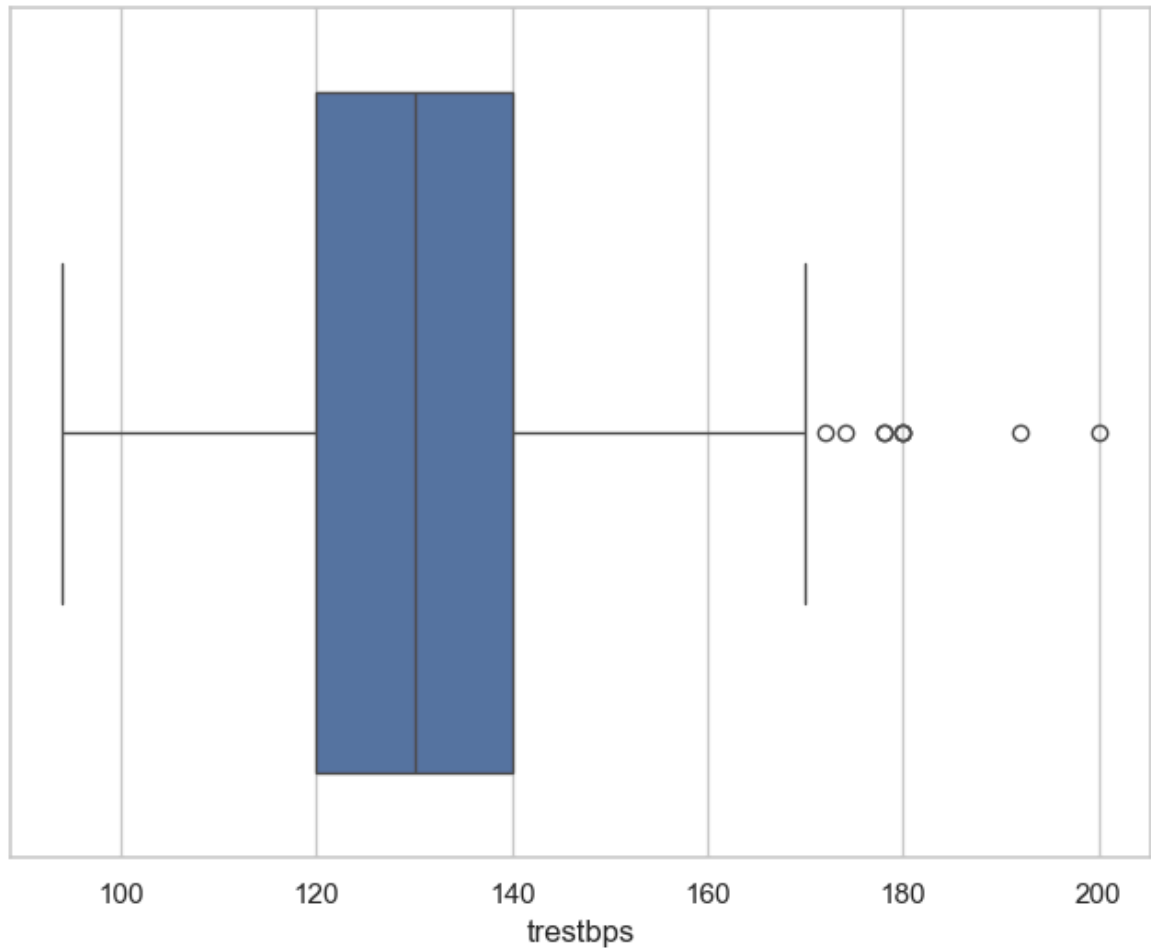
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["age"])
plt.show()
```

```
In [69]: df['trestbps'].describe()
```

```
Out[69]: count    303.000000
mean      131.623762
std       17.538143
min       94.000000
25%      120.000000
50%      130.000000
75%      140.000000
max      200.000000
Name: trestbps, dtype: float64
```

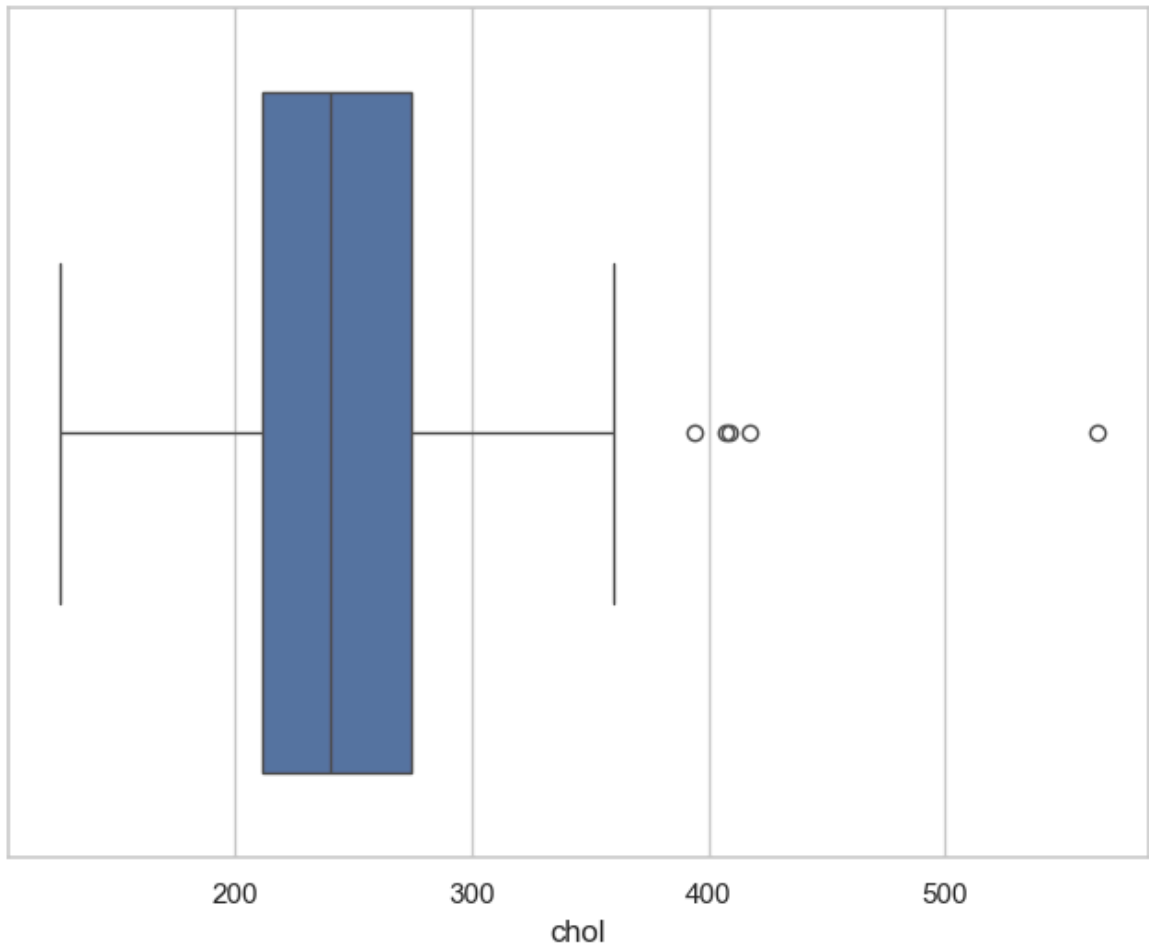
```
In [70]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["trestbps"])
plt.show()
```



```
In [71]: df['chol'].describe()
```

```
Out[71]: count    303.000000  
mean      246.264026  
std       51.830751  
min       126.000000  
25%      211.000000  
50%      240.000000  
75%      274.500000  
max       564.000000  
Name: chol, dtype: float64
```

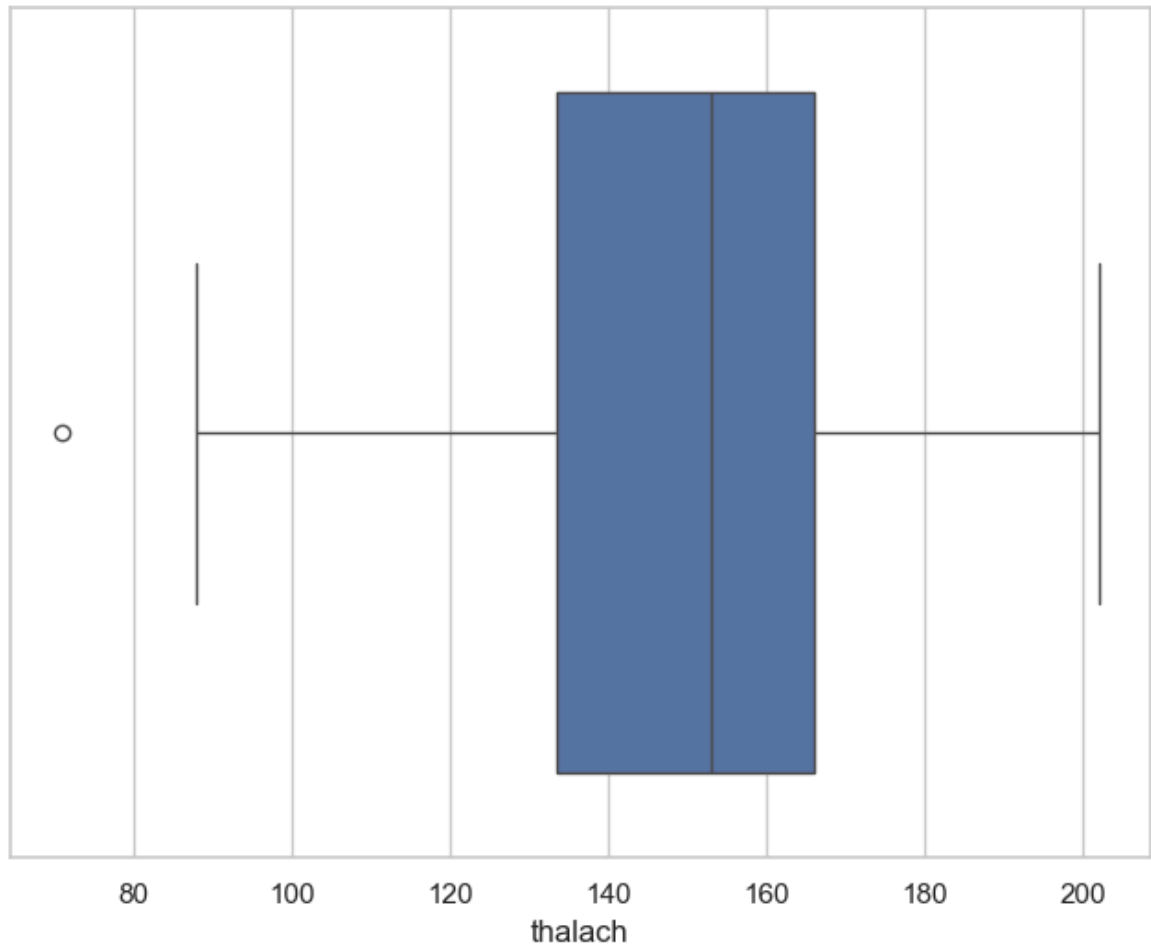
```
In [72]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["chol"])  
plt.show()
```



```
In [73]: df['thalach'].describe()
```

```
Out[73]: count    303.000000
mean      149.646865
std       22.905161
min       71.000000
25%      133.500000
50%      153.000000
75%      166.000000
max       202.000000
Name: thalach, dtype: float64
```

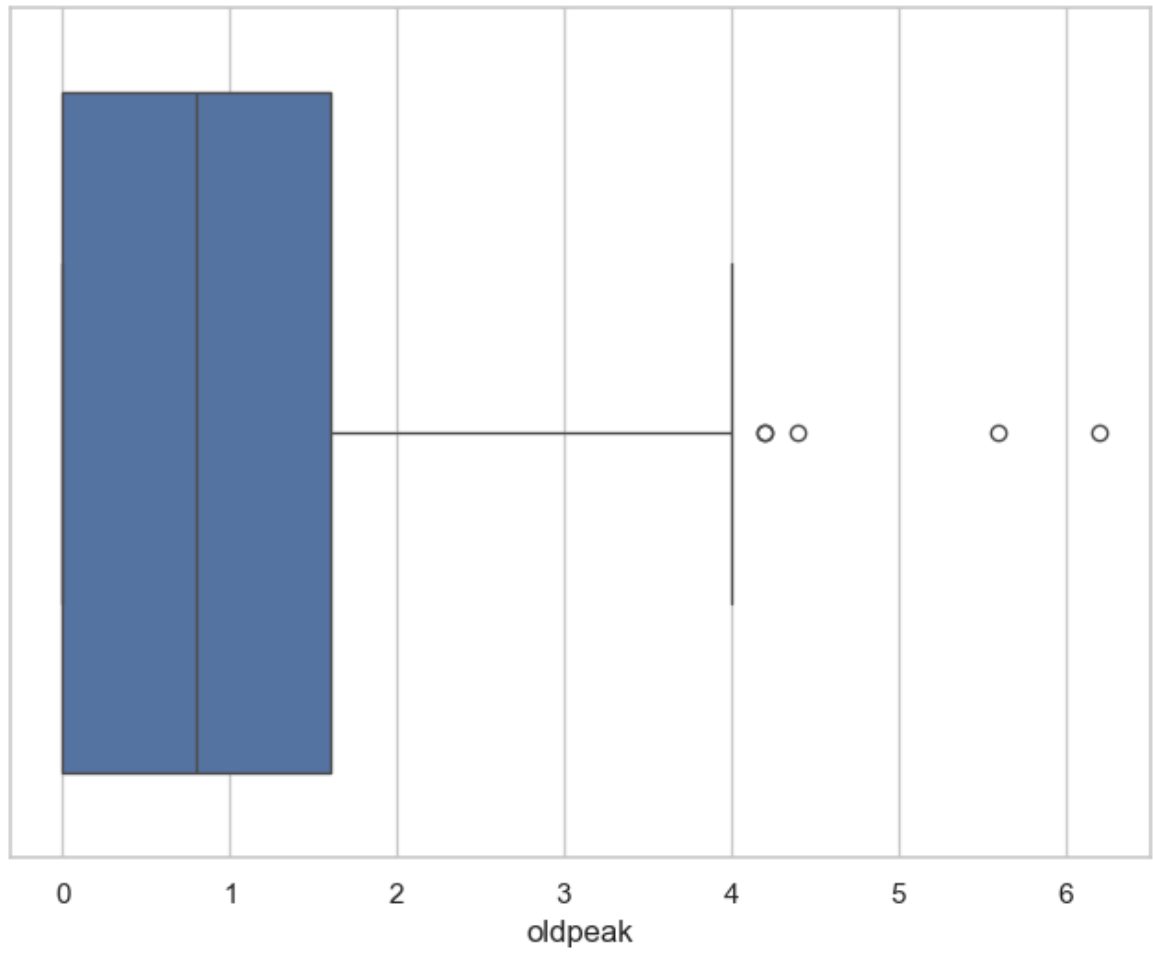
```
In [74]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["thalach"])
plt.show()
```



```
In [75]: df['oldpeak'].describe()
```

```
Out[75]: count    303.000000  
mean         1.039604  
std          1.161075  
min           0.000000  
25%           0.000000  
50%           0.800000  
75%           1.600000  
max           6.200000  
Name: oldpeak, dtype: float64
```

```
In [76]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["oldpeak"])  
plt.show()
```



In []: